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Accelerating Smart Production: Managing Sustainable Selection and Integration of Digital Technologies

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ACCELERATING SMART PRODUCTION

MANAGING SUSTAINABLE SELECTION AND INTEGRATION OF DIGITAL TECHNOLOGIES

Natalie Agerskans

2026



Department of Engineering Sciences

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Abstract

The manufacturing industry is undergoing a digital transformation in which an increasing range of digital technologies is available to support the development of smart production. However, realizing the potential of digital technologies depends on not only selecting suitable digital technologies, but also integrating them with existing production equipment, systems, processes, and people. Manufacturing companies commonly approach digital technology selection and integration through fragmented and technology-driven initiatives, which makes it difficult to establish coherent data flows, from raw data to usage, and realize the intended value over time. Moreover, production systems differ in their characteristics and needs; as a result, digital technologies suitable in one context may not be as suitable in another. The large number of digital technologies available also makes it challenging for manufacturing companies to determine which digital technologies to select and how to integrate them, which can lead to unsustainable selection and integration processes that, for instance, are resource-demanding or fail to deliver the intended value.

Therefore, the purpose of this PhD thesis is to develop guidance for managing the sustainable selection and integration of digital technologies to enable smart production. This thesis adopts a case study design to extract data from cases within different production system contexts.

The thesis shows that the production system context influences sustainable digital technology selection and integration through three mechanisms. First, it shapes requirements on how data need to be generated, communicated, stored, processed, and used to support decision-making. Second, contextual conditions and priorities influence which digital technology alternatives are feasible and what needs to be emphasized during selection and integration. Third, the production system context influences which selection and integration practices require emphasis and how they need to be applied. The findings also demonstrate that sustainable digital technology selection and integration is an iterative process consisting of several phases: prepare, select, integrate, and ensure value. Throughout these phases, data value-chain requirements, contextual conditions and priorities, and selection and integration practices need to be considered differently depending on the phase and production system context. The processes highlight the importance of continuous management of competing demands and adaptations based on learnings throughout the process. The process provides manufacturing companies with guidance for moving from fragmented digital technology initiatives toward a more systematic and adaptive approach to the selection and integration of digital technologies. Such guidance can support manufacturing companies in accelerating smart production.

Sammanfattning

Tillverkningsindustrin genomgår en digital transformation där ett allt större utbud av digitala teknologier finns tillgängliga för att stödja utvecklingen av smart produktion. För att realisera potentialen i digitala teknologier krävs dock inte bara att lämpliga digitala teknologier väljs, utan också att de integreras med befintlig produktionsutrustning, system, processer och människor. Tillverkningsföretag hanterar ofta val och integration genom fragmenterade och teknikdrivna initiativ, vilket försvårar sammanhängande dataflöden och realisering av avsett värde över tid. Produktionssystem skiljer sig dessutom åt vad gäller egenskaper, behov och kontextuella förutsättningar, vilket innebär att digitala teknologier som är lämpliga i ett visst produktionssystemkontext inte nödvändigtvis passar i ett annat. Det stora utbudet av digitala teknologier på marknaden gör det även svårt att avgöra vilka som bör väljas och hur de ska integreras. Detta kan leda till resurskrävande processer för val och integration som inte resulterar i avsett värde.

Syftet med denna doktorsavhandling är därför att utveckla vägledning för hur val och integration av digitala teknologier kan hanteras hållbart för att möjliggöra smart produktion. Avhandlingen bygger på fallstudier i olika produktionssystemkontexter.

Avhandlingen visar att produktionssystemkontexten påverkar hur val och integration behöver hanteras genom tre mekanismer. För det första formar produktionssystemkontexten kraven längs datavärdekedjan, det vill säga hur data behöver genereras, kommuniceras, lagras, bearbetas och användas för att stödja beslutsfattande. För det andra påverkar kontextuella förutsättningar och prioriteringar vilka digitala teknologier som är genomförbara och vad som behöver betonas under val och integration. För det tredje påverkar produktionssystemkontexten vilka arbetssätt som behöver betonas och hur de behöver tillämpas. Resultaten visar även att val och integration behöver hanteras som en iterativ process bestående av fyra faser: förbereda, välja, integrera och säkerställa värde. Genom faserna behöver krav kopplade till datavärdekedjan, kontextuella förutsättningar och prioriteringar samt arbetssätt för val och integration beaktas olika beroende på fas och produktionssystemkontext. Processen belyser även vikten av att kontinuerligt hantera konkurrerande krav och anpassa arbetssättet utifrån lärdomar under processens gång. Sammantaget ger avhandlingen vägledning för hur tillverkningsföretag kan hantera val och integration av digitala teknologier på ett systematiskt och kontextanpassat sätt för att stödja en hållbar och accelererad utveckling mot smart produktion.

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To everyone who has been part of this journey: thank you.

Natalie

Publications

Appended Papers in the Thesis

This thesis is based on the following papers, which are referred to in the text by their roman numerals.

I. Agerskans, N., Ashjaei, M., Bruch, J., Chirumalla, K. (2023) Critical Factors for Selecting and Integrating Digital Technologies to Enable Smart Production: A Data Value Chain Perspective. Advances in Production Management Systems. IFIP International Conference on Advances in Production Management Systems, https://doi.org/10.1007/978-3-031-43662-8_23

II. Agerskans, N., Ashjaei, M., Bruch, J., & Chirumalla, K. (2025). A data flow framework to support the selection and integration of digital technologies for smart production. International Journal of Production Research, 1-18. https://doi.org/10.1080/00207543.2024.2447931

III. Agerskans, N., Bruch, J., Chirumalla, K., Ashjaei, M. Practices for Selecting and Integrating Digital Technologies Across the Data Value Chain. Under review for the journal Production & Manufacturing Research.

IV. Agerskans, N., Bruch, J., Ashjaei, M., Leberruyer, N., Chirumalla, K., Navigating Contextual Complexity: A Comparative Study on the Selection and Integration of Digital Technologies in Smart and Sustainable Production. Procedia Computer Science, 277, 705-717. https://doi.org/https://doi.org/10.1016/j.procs.2026.02.112

V. Towards Smart and Sustainable Production: A Conceptual Framework for Sustainability Considerations in Digital Technology Selection and Integration. IOP Conference Series: Materials Science and Engineering, 1342, 012041. https://doi.org/10.1088/1757-899X/1342/1/012041

VI. Agerskans, N., Bruch, J., Chirumalla, K., Ashjaei, M. Managing Tensions in Digital Technology Selection and Integration for Smart Production: A Global Manufacturing Perspective. Under review for the International Journal of Production Research.

Together with my supervisors Jessica Bruch, Mohammad Ashjaei, and Koteshwar Chirumalla, I identified and formulated the research problem. I planned the research, collected and analyzed data, and wrote and revised the manuscript after receiving feedback from Jessica Bruch, Mohammad Ashjaei, and Koteshwar Chirumalla.

Additional Publications

Agerskans, N., Bruch, J., Chirumalla, K., Ashjaei, M. (2022) Enabling Smart Production: The Role of Data Value Chain. Advances in Production Management Systems. Smart Manufacturing and Logistics Systems: Turning Ideas into Action. Cham: Springer Nature Switzerland, pp. 477-485, https://doi.org/10.1007/978-3-031-16411-8_55

Agerskans, N. (2023). Digital Technologies for Enabling Smart Production Examining the Aspects of Selection and Integration. Licentiate thesis. Mälardalen University (Sweden)

Agerskans, N., Ahlskog, M., Badasjane, V., Granlund, A., & Sauter, B. (2025). Coordination of Digital Transformation. In: Mälardalens universitet.

Agerskans, N., & Leberruyer, N. (2026). Data Needs in Cross-Functional Remanufacturing Decisions: Insights from Engineering, Sales, and Production. IOP Conference Series: Materials Science and Engineering, 1342, 012026. https://doi.org/10.1088/1757-899X/1342/1/012026

Berrouyne, I., Govindaraj, A., & Agerskans, N. (2026). Perceived Complexity in Automation Systems: Sources, Impacts, and Perspectives from Industry. Procedia Computer Science, 277, 1476-1486. https://doi.org/https://doi.org/10.1016/j.procs.2026.02.186

Fongang, E., Mahmood, D., Jackson, M., Bellgran, M., & Agerskans, N. (2026). An IIoT based Approach to monitoring production performance-A case study. IOP Conference Series: Materials Science and Engineering, 1342, 012038. https://doi.org/10.1088/1757-899X/1342/1/012038

Leberruyer, N., Agerskans, N., Elvin, M., & Chirumalla, K. (2024). Remanufacturing Components Using Twin Transition–An Exploratory Study in the Heavy-Duty Vehicle Industry. 11th Swedish Production Symposium (SPS2024), Trollhättan from 23 to 26 April 2024

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1 Introduction

This chapter describes the importance of the research area, introduces digital technologies in production systems, and outlines the need for a sustainable approach to managing their selection and integration to enable smart production. The research problem is defined, and the purpose and two research questions are presented. The chapter ends by describing the scope and outline of this thesis.

1.1 Background

Production can be viewed as a system in which people, processes, production equipment, information systems, and other elements are organized to transform raw materials into finished products (Bellgran & Säfsten, 2010). The ongoing digital transformation is changing how production systems operate. By using digital technologies to interconnect elements within the production system, manufacturing companies can generate valuable insights about their production processes, enabling greater awareness and fact-based decision-making (Colombari et al., 2024; Schuh et al., 2020a). Digital technologies include a wide range of software and hardware technologies that enable the digitalized and automated handling of data (Berger et al., 2018; Ghobakhloo, 2020). Examples of digital technologies include cloud computing, the Industrial Internet of Things (IIoT), artificial intelligence, and big data analytics (Rajković et al., 2025; Zheng et al., 2020). In this thesis, smart production is understood as a production system in which digital technologies are used to manage data to support fact-based decisions and actions.

Digital technologies can be used for different purposes within a production system, such as supporting production planning, predictive maintenance, quality control, and material handling (Zheng et al., 2020). Yet different purposes require different types of data for decision support, such as production status, error codes, and machine stops (Freitas et al., 2026). Moreover, production systems differ in their characteristics, including their production processes, equipment, information systems, people, organizational structures, and available resources, which create different conditions for digital technology initiatives (Yli-Viitala & Helo, 2026). For example, small and medium-sized enterprises (SMEs)

commonly have limited resources whereas large manufacturing companies usually have more access to specialized competences (Wu et al., 2024). Differences in production equipment also affect how data can be generated, as older production equipment can have limited connectivity and interoperability, which makes automated data collection more difficult than with newer equipment (Keshav Kolla et al., 2022). Similarly, differences in the end-users of data affect how data need to be visualized and presented (Lindner et al., 2025). For example, operators may require more detailed real-time data to support immediate actions while production managers may require aggregated performance data to support higher-level decisions (Böck et al., 2025). Consequently, how manufacturing companies develop toward smart production should be influenced by the specific production system context.

An important part of developing toward smart production is the selection and integration of digital technologies. The selection of digital technologies refers to identifying and evaluating technology alternatives to determine which technologies best support organizational needs, capabilities, and objectives (Reza Hamzeh et al., 2018; Maretto et al., 2025; Santos et al., 2023). Integration refers to embedding selected digital technologies into the existing production system and adapting them to its people, processes, production equipment, information systems, and other elements (Ortega et al., 2022; Tortorella et al., 2021). The selection and integration of digital technologies are closely related activities (Silva et al., 2022). For example, compatibility with existing digital technologies and systems should be considered during technology selection (Aballay et al., 2023).

As production systems differ, no single digital technology or integration approach is suitable for all production system contexts. Therefore, the selection and integration of digital technologies need to be based on the specific production system in which they will be used, including considering how digital technologies fit with existing production system elements and if they can support the intended purpose.

1.2 Problem Formulation

Despite the demonstrated potential of digital technologies in research, manufacturing companies commonly face difficulties in selecting and integrating them into practice (Eichenseer & Winkler, 2024). Such difficulties can be explained by several underlying challenges. First, digital technologies develop rapidly. With more than 60 overall types of digital technologies already identified and new technologies continuously emerging (Klingenberg et al., 2019; Liu & Zolghadri, 2026), manufacturing companies face uncertainties about which digital technologies to select. As digital technologies can be applied for different

purposes, such as planning, logistics, maintenance, and quality control (Tao et al., 2018), their suitability depends on the intended purpose and the production system context in which they will be used (Martín-Gómez et al., 2024). Moreover, digital technologies do not create value in isolation, as their functionality depends on their integration and interconnection with other technologies (Varriale et al., 2025). Manufacturing companies therefore need to consider how combined digital technologies support the intended purpose (Klingenberg et al., 2019). Indeed, the inappropriate selection and integration of digital technologies can consume substantial resources without generating the intended value (Horváth & Szabó, 2019; Sjödin et al., 2018). In addition, realizing and sustaining value through digital technologies' integration remain challenging, as such activity requires alignment with existing resources, capabilities, and ways of working (Franzè et al., 2024). Therefore, in this thesis, sustainable selection and integration refer to selecting and integrating digital technologies in ways that enable their intended value to be realized and sustained over time.

Achieving the sustainable selection and integration of digital technologies is further complicated by differences between production system contexts. Previous research on digital technology selection has predominantly focused on quantitative decision-support approaches, including multi-criteria decision-making methods and assessment criteria for evaluating and comparing technology alternatives (e.g., Aballay et al., 2023; Maretto et al., 2022; Maretto et al., 2025; Santos et al., 2023). Meanwhile, research on digital technology integration has instead focused on identifying determinants, barriers, critical success factors, and integration frameworks (Bueno et al., 2023; Ghobakhloo, 2019, 2020; Kassem et al., 2024). Although these studies consider different aspects of the production system context, these aspects are often considered separately rather than assessing the production system as a whole. Yet considering these aspects separately can have a negative impact as smart production creates interdependencies among technologies, organizational structures, processes, and individual capabilities (Colombari et al., 2024). Consequently, sustainable selection and integration require the consideration of technical and social production system elements together rather than independently (Sony & Naik, 2020). Thus, considering individual contextual aspects or technologies separately may overlook dependencies that affect whether a technology can be successfully integrated and generate its intended value.

Moreover, manufacturing companies face challenges in managing the selection and integration of digital technologies throughout the process. Digital technology initiatives can involve multiple organizational functions, particularly operational technology (OT) and information technology (IT), which need to coordinate different responsibilities, competences, and requirements (Massaro Mendes da Silva & Albertin, 2024; Vaya-Arboledas et al., 2025). Unclear governance

structures and limited alignment between strategic objectives and operational needs can also constrain the coordination of digital technology initiatives, leading to initiatives being pursued as isolated projects rather than as part of a coherent development toward smart production (Bertoncel et al., 2018; Santos et al., 2023).

Although some studies consider both the selection and integration of digital technologies (Silva et al., 2022), previous research has predominantly approached selection through technology evaluation and decision-support method and has separated integration as the focus on integration factors and frameworks. However, decisions made when selecting digital technologies create requirements for their subsequent integration. At the same time, challenges encountered during integration may require previous decisions to be reconsidered. Managing selection and integration therefore requires coordination across activities, organizational functions, and production system elements rather than treating technology selection as a decision followed by a separate integration process. The need for continuous management over time is further reinforced by the rapid evolution and shorter lifecycles of digital technologies, which can require technologies to be updated, replaced, or reintegrated (Liu & Zolghadri, 2026).

Taken together, previous research provides valuable knowledge on individual aspects of the selection and integration of digital technologies. However, limited guidance is provided on how these activities should be managed together while accounting for the production system context and changes that emerge over time. Therefore, guidance on how manufacturing companies can manage the sustainable selection and integration of digital technologies to enable smart production is needed.

1.3 Purpose and Research Questions

Against the background and problem formulation described thus far, the **purpose** of this PhD thesis is to develop guidance for managing the sustainable selection and integration of digital technologies to enable smart production. This thesis is intended to support manufacturing companies in orchestrating resources to accelerate progress toward smart production.

Production systems differ in terms of their characteristics, resources, capabilities, and processes, which can affect both the suitability of different digital technologies and how they should be integrated to generate value. As such, understanding the ways in which the production system context influences the sustainable selection and integration of digital technologies is important for ensuring that these

digital technologies generate value over time. This context leads to the first research question:

RQ1 - How does the context of the production system influence the sustainable selection and integration of digital technologies for smart production?

Understanding the influence of the production system's context provides a basis for the sustainable selection and integration of digital technologies. However, guidance is needed to help manufacturing companies approach their selection and integration in a sustainable manner. This need leads to the second research question:

RQ2 - How should the sustainable selection and integration of digital technologies be managed in a production system context?

1.4 Scope of the PhD Thesis

The purpose of this PhD thesis is to develop guidance to manage the sustainable selection and integration of digital technologies to enable smart production. The scope of the thesis is delimited by the concepts of production system context, smart production, and the sustainable selection and integration of digital technologies.

The production system constitutes the context in which the selection and integration of digital technologies are studied. The primary focus is on activities related to the transformation of materials into components or products, including activities such as machining and assembly. It also includes considerations for people, processes, production equipment, information systems, organizational structures, and resources associated with these activities. From a smart production perspective, the thesis focuses on how digital technologies support production-related decision-making and actions. However, the scope is not limited to digital technologies in the production system as a production decision could require the use of digital technologies to collect data from outside the production system boundaries.

Within the sustainable selection and integration of digital technologies, the focus is not on developing new technologies, but on how existing digital technologies can be evaluated, selected, combined, and integrated into production systems. This approach includes considering their suitability for the intended purpose and production system context as well as how they can be integrated with existing production system elements. Furthermore, the selection and integration of digital technologies are not considered from a higher strategic perspective, but rather on

cross-functional teams (including IT/OT competence, production managers, project managers, engineers, and operators) that work with smart production.

This thesis also covers case studies in several application areas, including machine monitoring, remanufacturing, quality control, material handling, and production planning. Two SMEs and two large global manufacturing companies are included. The case studies focus on selecting and integrating digital technologies into production systems to support specific decisions. Together, these cases provide insights into different production system contexts, real-world challenges, and practices related to the sustainable selection and integration of digital technologies.

1.5 Outline of the Thesis

The remainder of this thesis is organized as follows. Chapter 2 presents the theoretical background used in this thesis. Chapter 3 details the research methodology. Chapter 4 summarizes the appended papers. Chapter 5 presents the results of this work by analyzing and synthesizing the findings related to the two research questions. Finally, Chapter 6 discusses the findings, contributions, and concludes this thesis. The six papers underlying this thesis are appended to this work.

2 Theoretical Framework

This chapter presents the theoretical framework for the thesis. It introduces production systems and different production system contexts, followed by smart production and the role of digital technologies. Finally, the chapter addresses the sustainable selection and integration of digital technologies.

2.1 Production Systems

2.1.1 Defining Production Systems

The terms *manufacturing* and *production* are often used as synonyms. However, in this research, these terms are distinguished as argued for in the past (Bellgran & Säfsten, 2010; Bruch, 2012; Rösiö, 2012). In this argument, manufacturing is seen as hierarchical superior production, whereas production refers to the physical process of making a product from raw materials. Manufacturing involves a series of interrelated operations and activities, including production, as well as additional functions such as product design, shipping, and sales (Bellgran & Säfsten, 2010).

Production requires a combination of many different elements to transform raw materials into finished products. These elements include, for instance, people, processes, production equipment, systems, and networks. As the relationship among all these elements is often quite complex, a system perspective can be applied (Bellgran & Säfsten, 2010). A system can be defined as "a collection of different components, such as for example people and machines, which are interrelated in an organised way and work together towards a purposeful goal" (p. 38). Thus, the production system can be seen as a limited part of the manufacturing system that denotes the activities and elements needed to transform raw materials into finished products or parts of products. Compared to the production system, the manufacturing system considers all activities and elements needed to put products on the market (Bellgran & Säfsten, 2010; Bruch, 2012; Rösiö, 2012). Furthermore, the production system consists of several subsystems that can include both parts production and assembly. Figure 1 depicts a hierarchical perspective of the production system.

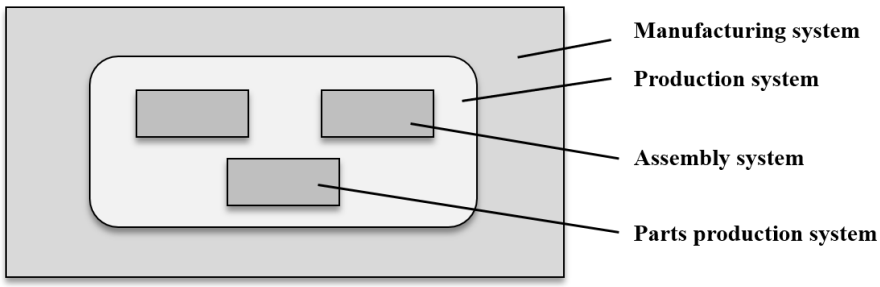


Figure 1: A hierarchical perspective on production system (Bellgran & Säfsten, 2010)

Increasing product variety and shorter product lifecycles have created new demands on production system (Hu et al., 2011). To meet these challenges and stay competitive, manufacturing companies need to constantly improve their production systems by making changes (Bellgran & Säfsten, 2010; Holweg, 2008). Such changes may include the use of new technology enabling the production of new products via more efficient production processes (Bellgran & Säfsten, 2010). Changes made in one part of the production system will have an impact on subsystems as well as processes (Gopalakrishnan et al., 1999). Therefore, it is important to carefully consider all elements within the system as well as their relationships in order to enable a production system that creates value for the customer (Aslam et al., 2018; Bellgran & Säfsten, 2010).

2.1.2 Production System Contexts

Production systems can exist in different contexts, which vary in terms of their organizational boundaries, structure, and activities performed within the system. One distinction concerns the size and organizational structure of the manufacturing company. Compared with large manufacturing companies, SMEs are generally characterized by flatter and less formal organizational structures, fewer specialized resources, and individuals assuming responsibilities across several functions (Cragg et al., 2011). Organizational structure can differ in dimensions such as the number of hierarchical levels, degree of functional specialization and integration, distribution of decision-making responsibilities, and degree of formalization (Nahm et al., 2003). As organizations increase in size and activities become distributed across organizational units and production sites, then the production system may consequently involve a greater number of organizational interfaces and coordination needs.

Another production system context is an international manufacturing network (IMN), where production is distributed across multiple production sites in

different geographical locations that are interconnected as part of a larger manufacturing system
(Ferdows, 1997; Shi & Gregory, 1998).

In global manufacturing companies, a production system can be distributed across several production sites in different geographical locations. These sites together form an international manufacturing network (IMM) that needs to be coordinated as part of the larger production system (Cheng & Farooq, 2018; Shi & Gregory, 1998). Production sites within the same manufacturing company can differ in their products, production processes, resources, competencies, responsibilities, and level of autonomy (Cheng & Farooq, 2018). Coordination is therefore needed both within individual production sites and across several sites. This can include sharing information and knowledge, coordinate production activities and resources, and develop common ways of working across the global organization (Badasjane et al., 2022; Rudberg & Martin West, 2008; Vereecke et al., 2006).

Production system contexts can also differ depending on the application area under consideration. A production system comprises several interrelated activities with different purposes and characteristics, such as production planning, production monitoring, product quality control, and maintenance, (Bellgran & Säfsten, 2010; Tao et al., 2018). These application areas are interrelated, as changes or disturbances in one area can influence activities in others (Colledani et al., 2014; Farahani & Tohidi, 2021; Gopalakrishnan et al., 1999).

Different application areas also require different types of data and decision support. Depending on the purpose, decision support can range from providing an understanding of the current state, to explaining why something has happened, anticipating future developments, and supporting decisions on how to respond (Schuh et al., 2020a). For example, production monitoring focuses on understanding production performance and identifying deviations that require action (Reuter & Brambring, 2016), whereas production planning concerns future production requirements and how production resources and processes should be planned (Valencia et al., 2019). Maintenance requires information about the current condition of production equipment as well as support for anticipating failures (Jardine et al., 2006), whereas product quality control focuses on identifying deviations from product quality goals and taking corrective actions (Juran, 1999). Thus, the data and decision support needed within a production system depends on the purpose and characteristics of the application area.

Consequently, the production system context influences how the production system is organized and coordinated, as well as the data and decision support needed for different application areas.

2.2 Smart Production

Smart production is driven by the opportunities of using digital technologies to manage data, one of today's most valuable resources, to support informed decision-making and enable resilient, efficient, and competitive production systems (Kusiak, 2018; Maddikunta et al., 2022; Tao et al., 2018). By using digital technologies, production equipment, systems, and people can be interconnected, enabling data to be generated, shared, and used in real time (Kusiak, 2018; Schlemitz & Mezhuyev, 2024). This allows better visibility of production processes and supports fact-based and timely decision-making (Helu et al., 2016; Schuh et al., 2020a). For instance, connected machines can communicate with operators, systems, and other machines to detect problems, support problem solving, and improve production performance (Eichenseer & Winkler, 2024; Tao et al., 2018). As a result, smart production can be understood as a production system in which digital technologies are used to manage data and support fact-based decisions and actions, with the aim of improving the performance and adaptability of the production system (Eichenseer & Winkler, 2024; Kusiak, 2018; Tao et al., 2018).

2.2.1 Digital Technologies for Smart Production

Digital technologies include both hardware and software technologies that enable a digitalized and automated handling of data to improve process efficiency and enhance value-creating activities (Berger et al., 2018; Ghobakhloo, 2020; Parhi et al., 2022). These technologies include a wide variety of simpler to extremely complex digital technologies, where each one constitutes different capabilities (Klingenberg et al., 2019). Examples of digital technologies include cloud computing, IIoT, AI, augmented reality (AR), and simulation (Klingenberg et al., 2019; Xu et al., 2018; Zheng et al., 2020).

Digital technologies for enabling smart production rarely create value in isolation (Varriale et al., 2025). Instead, different technologies contribute to different activities to transform raw data into information that can support production activities and decision-making (Klingenberg et al., 2019; Tao et al., 2018). These activities can be understood through the data value chain, which describes the sequence of steps required to generate value from data (Cavanillas et al., 2016). As Figure 2 illustrates, the data value chain used in this thesis consists of five phases: data generation, data communication, data storage, data processing, and data usage:

- *Data generation*. Data generation includes data sources and data collection. Data sources refer to the points of origin where data are generated, and data generation is the method used for data acquisition (AlSuwaidan, 2021). Examples of data sources include products, production

equipment, and production processes (Tao et al., 2018). Production equipment with built-in data collection capabilities, sensors, and radio frequency identification (RFID) (AlSuwaidan, 2021; de Jesus Pacheco et al., 2023; Manimuthu et al., 2022) are examples of digital technologies that can be used for data collection.

- *Data communication.* Data communication refers to the transmission of data from their sources to the places where they will be stored, processed, and visualized. Various digital technologies based on wireless and wired communication fall within this category, including WiFi, ethernet, and 5G (Ghobakhloo, 2020; Rao et al., 2021). IIoT platforms are also commonly used for connecting different data sources via the internet (Arnold et al., 2021; Parhi et al., 2022).
- *Data storage*. As the name suggests, data storage is the place where data are stored for analysis or until used. Typical digital technology examples include databases and cloud storage (O'Donovan et al., 2019).
- *Data processing.* Data processing refers to the operations conducted to extract information from data, such as by using AI and big data analytics (Cioffi et al., 2020; Maddikunta et al., 2022; Manimuthu et al., 2022).
- *Data usage.* Data visualization and usage are the techniques used to present data to end-users and the context in which data are applied to add value. For example, relevant data can be visualized in the form of charts, diagrams, or figures using tablets, monitors, augmented reality, or virtual reality on the shop floor (Burova et al., 2022; Simões et al., 2018; Tao et al., 2018). IIoT platforms are also common technologies that serve this purpose (Arnold et al., 2021).



Figure 2: The five phases of the data value chain

Although the topic of digital technologies for smart production has **attracted** greater attention in recent years (Silva et al., 2022), research has predominantly focused on the functionality and value of individual technologies. Less focus has been on how multiple digital technologies work together as an interconnected system (Battaglia et al., 2023; Varriale et al., 2025). However, digital technologies provide complementary capabilities, and a combination of technologies enables data to be generated, communicated, stored, processed, and ultimately used to create value by data usage (Klingenberg et al., 2019; Tao et al., 2018). For example, sensors and connected production equipment may enable data

generation, but the generated data also need to be communicated, stored, processed, and presented in a form that can support production activities and decision-making (Klingenberg et al., 2019; Tao et al., 2018). Consequently, different digital technologies support different phases of the data value chain, and the functionality and value provided by one technology can depend on capabilities provided by other digital technologies (Battaglia et al., 2023; Varriale et al., 2025).

The phases of the data value chain should therefore not be viewed as independent activities, but as interconnected parts of an overall data flow (Bendlin et al., 2024; Tao et al., 2018). Limitations in one phase can reduce the value that can be achieved in subsequent phases. For instance, insufficient data quality during data generation can limit the usefulness of processing and therefore also decision-making (Goknil et al., 2023). Moreover, limitations in data communication can constrain the timely exchange and use of production data (Schlemitz & Mezhuyev, 2024). Similarly, processed data only creates value when the resulting data can support relevant decisions and actions (Bendlin et al., 2024). Thus, considering the complete data value chain provides a way to understand how different digital technologies complement and depend on each other in enabling smart production (Battaglia et al., 2023; Tao et al., 2018).

Without coordination across these phases, digital technologies risk becoming isolated solutions that provide local functionality without contributing to the overall objectives of smart production (Klingenberg et al., 2019; Tao et al., 2018). As such, the data value chain provides a perspective for understanding the technological interdependencies involved in transforming raw data into useful information and value.

2.2.2 Smart Production as a Socio-Technical System

Production systems can be viewed as socio-technical systems consisting of people, processes, and technologies. These elements are interrelated and need to be considered when enabling smart production.

The fundamental element in any manufacturing company is the people working there (Hozdić & Makovec, 2023). Integrating digital technologies can have a significant impact on the people working within the production system (Ciffolilli & Muscio, 2018), including operators working on the shop floor, engineers designing systems, and managers making strategic decisions (Saniuk et al., 2020; Sjödin et al., 2018; Zizic et al., 2022). It is, therefore, helpful if important stakeholders can adapt to new ways of working with digital technologies and new ways of learning and sharing data (Lepore et al., 2022). As Bellgran and Säfsten

(2010, p. 149) explained, “how, what, and in what way people think affects of course the final result, and not least—the way to get there.”

For example, a lack of involvement, an unclear purpose, or a low trust in data can lead to resistance to or the underutilization of technologies (Tortorella et al., 2020). Thus, smart production emphasizes human centricity by improving the quality of people’s working conditions and workplaces (Maddikunta et al., 2022), which includes improving ergonomics, such as eliminating heavy lifting and repetitive movements (Scalona et al., 2023), as well as supporting meaningful work competence development and well-being (Zizic et al., 2022). These aspects are closely linked to the use of digital technologies, as the usability, purpose, and integration of technologies influence how people interact with data in their daily work.

Another important element in smart production is the processes, referring to how production and organizational activities are organized. Processes also define how digital technologies are integrated, used, and maintained over time, linking technological capabilities with organizational ways of working. As Sharma et al. (2014) pointed out, “despite the hopes of many, insights do not emerge automatically out of mechanically applying analytical tools to data. Rather, insights emerge out of an active process of engagement between analysts and business managers using the data and analytic tools to uncover new knowledge.” (p. 435) Consequently, the need exists to consider the processes and ways of working in digital technology initiatives (Ghobakhloo, 2020; Schneider, 2018; Sjödin et al., 2018).

Yet, many work processes are changing. Today, organizational activities are characterized by decentralization, where decision-making processes are delegated by top managers to lower-level managers or sometimes to shopfloor workers (Molino et al., 2020). The objective of decentralization is to shorten the decision-making process by those who have the most information and ability to react in real time. However, to enable this way of working, data presented to decision-makers must provide useful information presented in a user-friendly way when needed to enable fast decisions and actions (Klingenberg et al., 2019; Schuh et al., 2020a).

The elements of people, process, and technology in smart production are closely interrelated and cannot be understood in isolation (Bellgran & Säfsten, 2010; Zizic et al., 2022). Instead, value is generated through the interactions among these elements. Digital technologies enable new ways of generating, processing, and using data, but their actual value is determined by how processes are shaped around them and how people use the data (Schuh et al., 2020a). This interdependence means that changes in one element will influence the others,

highlighting the complex nature of a production system (Gopalakrishnan et al., 1999). Similarly, Schuh et al. (2020b) emphasize that progress toward smart production requires coordinated development across resources, information systems, organizational structure, and culture. For example, new digital technologies can require new skills, work processes, and responsibilities.

2.3 Sustainable Selection and Integration of Digital Technologies

Selecting and integrating digital technologies are essential activities for manufacturing companies that want to achieve smart production. The selection phase involves deciding in which technologies to invest, while the integration phase is about making sure the chosen technologies work well within the existing production system.

2.3.1 Selection of Digital Technologies

In general, technology selection involves gathering information about different options and comparing them based on predefined criteria and technical requirements, such as cost, risk, how well they fit the organization, and how easily they can be accepted by users (e.g. operators, engineers, and managers; (R Hamzeh et al., 2018; Karsak & Ahiska, 2005; Lamb & Gregory, 1997). However, manufacturing companies often tend to choose technologies in an unstructured way. Instead of following a clear process, decision-makers may rely on intuition or be influenced by trends, which can lead to poor investments (Bertoncel et al., 2018; Santos et al., 2023). Consequently, a more sustainable approach to selecting digital technologies is needed, which means having a structured approach that reduces the risks of unprofitable investments (Chu et al., 2019; Silva et al., 2022; Sjödin et al., 2018). Selecting the right technology is challenging because new digital solutions are constantly evolving, and manufacturing companies may lack the internal knowledge needed to fully understand their benefits and how to use them. Without a clear purpose and requirements, there is a risk of selecting digital technologies that do not contribute to value creation (Ghobakhloo, 2020; Klingenberg et al., 2019; Sjödin et al., 2018). Moreover, digital technologies often rely on one another to deliver their full potential, making it insufficient to consider them in isolation (Klingenberg et al., 2019). Instead, it is essential to understand how they can work together to create synergistic effects; for example, although some digital technologies are well-suited for data generation, others excel in data analysis, visualization, or storage (Agerskans et al., 2025).

2.3.2 Integration of Digital Technologies

Integration refers to the process of embedding digital technology in the production system, which means connecting the technology to existing machines, systems, and processes and end-users using the technology as intended (Ortega et al., 2022; Tortorella et al., 2021). To make this work, a holistic view is needed, considering both technical and process-related aspects (Bokrantz et al., 2017; Zizic et al., 2022). However, it is not just about installing new technology, but also about enabling smarter ways of working through the better use of data (Klingenberg et al., 2019; Manimuthu et al., 2022). Sustainable technology integration (i.e., achieving desired benefits, maintaining and continuing to use the technology) often requires changes in different parts of the manufacturing company. For instance, integrating new digital technology may involve training employees, updating processes, and improving coordination within the organization and across the supply chain (Ghobakhloo, 2018; Zizic et al., 2022). This highlights the complex nature of a production system as changes in one part of the system can affect subprocesses and subsystems (Bellgran & Säfsten, 2010). If digital technologies are not integrated in a sustainable way, they may not deliver the expected benefits, leading to poor utilization of resources and missed opportunities for desired improvements.

2.3.3 Managing Sustainable Selection and Integration of Digital Technologies

Managing the selection and integration of digital technologies involves structuring and coordinating the activities required to move from an identified production need to a digital technology that is integrated and creates value in production. Previous research has provided different approaches for supporting this work, including selection methods, frameworks, roadmaps, maturity models, and processes that guide manufacturing companies through different activities related to the selection and integration of digital technologies (Dillinger et al., 2022; Elhusseiny & Crispim, 2024; R Hamzeh et al., 2018; Schuh et al., 2020a). These approaches provide different types of guidance for how manufacturing companies can structure their work and make decisions.

However, the selection and integration of digital technologies have often been addressed separately in previous research. Research on selection has commonly focused on how digital technologies can be evaluated and selected (R Hamzeh et al., 2018; Maretto et al., 2022; Santos et al., 2023); meanwhile, research on integration has focused on how digital technologies can be integrated into production systems (Frank et al., 2019; Kassem et al., 2024; Ojha et al., 2024). Yet the two activities are closely connected. Decisions made during selection influence the conditions for integration, as selected technologies need to fit existing

technologies and systems (Agerskans et al., 2025; Kassem et al., 2024). Furthermore, the development toward smart production can occur in steps, where digital technologies and organizational conditions are developed over successive stages rather than through a single transformation (Schuh et al., 2020a). Managing selection and integration therefore requires an iterative approach in which activities and decisions can be revisited. More broadly, managing digital technology initiatives requires moving beyond isolated initiatives toward more systematic ways of working. Schuh et al. (2020b), for example, argue that isolated digital technology prototypes are insufficient for progressing toward smart production and emphasize the need for organizational and cultural changes alongside technical development.

The management of selection and integration activities also needs to consider whether digital technology creates the intended value (Colli et al., 2021; Ghobakhloo, 2018; R Hamzeh et al., 2018). From a Lean perspective, realizing value from an improvement requires that it becomes embedded in established ways of working and is continuously evaluated and improved (Liker, 2003). Thus, integration is not completed simply when digital technology has been introduced into the production system. How digital technologies are combined across the data value chain and remain aligned with the production system influences whether the intended value can be achieved and sustained (Franzè et al., 2024; Klingenberg et al., 2019). Managing selection and integration consequently requires manufacturing companies to consider the process from the initial need and selection of digital technologies through integration and continued value creation. In the context of smart production, digital technologies also need to be considered in relation to the broader sustainable objectives in relation to a production system (Ghobakhloo et al., 2024).

2.4 Summary of Theoretical Framework

The theoretical framework outlined the current understanding of smart production and the sustainable selection and integration of digital technologies in different production system contexts. It showed that digital technologies are important for enabling smart production, but that realizing their intended value depends on how they are selected and integrated with the existing production system. In particular, a deeper understanding is needed of how the production system context influences the selection and integration of digital technologies and how these activities can be managed to enable continued value creation. Three streams of literature inform this understanding: production systems and their contexts, smart production from a data value chain and socio-technical perspective, and the sustainable selection and integration of digital technologies.

3 Research Methodology

This chapter presents the research methodology used for this thesis. The chapter includes a description of the research method and approach, research process, research design, description of cases and data collection, and data analysis. Finally, the chapter ends with a reflection on the quality of this thesis.

My research process for the PhD studies can be divided into two phases: before and after the licentiate thesis defense. The first phase of the research process focused on examining the selection and integration of digital technologies to enable smart production, with an emphasis on key challenges and critical factors. The main conclusion from the licentiate thesis was that achieving smart production requires combining multiple digital technologies based on their roles across the data value chain (generation, communication, storage, processing, and usage). This conclusion implies that selection and integration should be guided by the specific data-related function each technology fulfills. However, digital technology alone is insufficient. Realizing value also requires considering people (e.g., change management and human involvement), processes (e.g., production and organizational workflows), and technology aspects (e.g., system modifications and integration requirements).

Together, these insights provide a broad overall understanding of what needs to be considered when selecting and integrating digital technologies for smart production. However, as this research was conducted based on the assumption that digital technologies need to be selected and integrated according to the requirements and conditions of the specific context (Klingenberg et al., 2019), it was necessary for the second phase to further investigate how different contextual conditions influence the selection and integration of digital technologies. At the same time, the scope needed to be narrowed for conducting more in-depth descriptive studies on how manufacturing companies can systematically manage the selection and integration of digital technologies, as not all potential contexts for smart production could be studied. Here, the choice was made to focus on sustainability. Therefore, the second phase sought to build further on the results from the licentiate thesis, but directed attention toward how the selection and integration of digital technologies can be made more sustainable.

3.1 Research Method

A case study method was selected to fulfill the purpose of this thesis. As Barratt et al. (2011) emphasized, it is important to clearly explain the rationale for choosing a case study method. In this research, the use of a case study method was motivated by three main reasons.

First, it was necessary to collect empirically rich data to ensure the practical relevance of the findings (Barratt et al., 2011). The selection and integration of digital technologies are activities that practitioners in the manufacturing industry carry out, and prior empirical research in this area is limited (Müller et al., 2018; Silva et al., 2022). Therefore, it was considered important to study the selection and integration of digital technologies in a real industrial setting. A case study method enables the collection of data grounded in the context and experiences of practitioners, thereby supporting the connection between academic insights and practical application (Fisher, 2007).

Second, this thesis addresses a complex and relatively underexplored contemporary phenomenon, where multiple interrelated factors need to be considered. A case study method is particularly suitable for in-depth and holistic understanding of such phenomena (Karlsson, 2016; Säfsten & Gustavsson, 2020; Voss et al., 2002). As this thesis investigates contemporary phenomena in real-life contexts when existing theoretical understanding is limited, a case study method is suitable (Yin, 2018).

Third, a case study method allows for studying a phenomenon over time (Yin, 2018), which is particularly relevant for this thesis, as it considers both the selection and integration of digital technologies—activities that typically occur in different phases. The integration of digital technologies can also be divided over longer time periods, such as starting with a pre-study before scaling up. Studying the selection and integration of digital technologies at a single point in time would not capture the dynamic and evolving nature. Considering this context, the case study method allows for a more comprehensive understanding of how these activities unfold (Karlsson, 2016; Yin, 2018).

3.2 Research Approach

As explained in the beginning of Section 3, this research process can be divided into two phases: before and after the licentiate thesis defense. The purpose of the PhD thesis, representing the second phase of the PhD process, is to provide guidance to manage the sustainable selection and integration of digital technologies to enable smart production. This thesis is intended to support manufacturing

companies in orchestrating resources to accelerate progress toward smart production. Thus, the practical relevance of research findings is considered essential. To enable this, a further conclusion from the first phase concerned the need for closer interactions with the manufacturing industry.

The very first case studied in the PhD process included close interactions and a weekly presence at a manufacturing company, which enabled deeper insights into practices and challenges. In contrast, later cases in the first phase—conducted during and after the Covid-19 pandemic—involved less frequent interaction, which limited the depth of understanding. This distinction indicated that more regular interaction was important for capturing detailed insights and strengthening practical applicability.

To increase practical relevance, an interactive research approach was adopted in the second phase, enabling a close collaboration with manufacturing companies (Ellström, 2008). Interactive research is characterized by collaboration between research and practice and aims at a joint learning and knowledge generation between the researcher and practitioners (Berglund et al., 2020; Ellström et al., 2020; Svensson et al., 2007). This research approach was applied through discussions with managers at different levels and organizational functions to develop a shared understanding of which problems to address. Thus, the process of framing the research focus and research design has been iterative and carried out in collaboration with managers (Pasmore et al., 2007; Shani et al., 2008). During the research process, close interactions with managers also enabled the continuous discussion of the findings.

3.3 Research Design

The research design guides the researcher in how to collect, analyze, and interpret data in a case study method. It acts as a plan on how to go from developing the research purpose and research questions to answering the research questions and drawing conclusions (Yin, 2018). Based on the conclusions from the licentiate thesis defense and the literature review, the initial research questions were formulated. Developing an initial understanding of the general construct and how the research questions can be structured for analysis is a prerequisite when using a case study method (Voss et al., 2002). As the research progressed, the research questions also evolved. This evolution is a natural part of the research process, as new insights emerge. However, the overall purpose of the study remained unchanged, and only minor adjustments were made to the research design.

3.3.1 Number of Cases

This research is based on the assumption that the sustainable selection and integration of digital technologies for smart production are strongly influenced by contextual factors that include, for instance, production characteristics, organizational structures, levels of digital maturity, and sustainability goals. Consequently, rather than there being one universally optimal way of selecting and integrating digital technologies, manufacturing companies need to adapt their approaches to their specific contexts and maturity levels.

To capture this contextual variation, a multiple case study design was applied. Studying multiple cases enables comparison across different production environments and maturity stages. These reduced risks that are commonly associated with single-case research, such as misinterpreting isolated events or limiting the generalizability of results (Voss et al., 2002; Yin, 2018).

At the same time, this thesis also includes a longitudinal single case study followed over more than 5 years. The single case study enabled an in-depth understanding of the selection and integration processes over time, capturing how decisions, challenges, and outcomes had evolved over time. While the multiple case studies provided data collection from different contexts, the longitudinal case study allowed for studying how processes develop over time (Yin, 2018). In addition, the single case study was conducted from an IMN perspective and included data collection from six manufacturing sites. This approach captured contextual variations across different production environments and organizational settings.

Figure 3 shows a timeline of the cases included in this thesis.

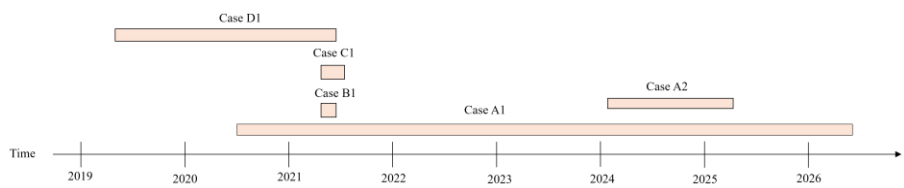


Figure 3: Timeline of the cases studied in this thesis

Table 1 presents an overview of the three case studies—Evoque, Velar, and Sport—and the cases included within each study.

Table 1: Overview of the number of cases in different case studies.

	Case study Evoque	Case study Velar	Case study Sport
Number of case(s)	4	2	1

3.3.2 Case Selection

The cases were selected using a theoretical sampling logic (Eisenhardt, 1989), meaning that cases were chosen based on criteria derived from theory and prior research (Säfsten & Gustavsson, 2020), rather than statistical representativeness. Following this logic, four criteria guided the selection of cases:
The cases were selected using a theoretical sampling logic (Eisenhardt, 1989), meaning that cases were chosen based on criteria derived from theory and prior research (Säfsten & Gustavsson, 2020), rather than statistical representativeness. Following this logic, four criteria guided the selection of cases:

1. **Variation of organizational context:** To ensure variation in organizational contexts, cases were selected from manufacturing companies operating in different industries, including both original equipment manufacturers (OEMs) and suppliers, and represented different company sizes. While variation is achieved across the overall set of companies, the inclusion of multiple cases in some studies and more in-depth analyses within Company A enables both cross-contextual comparisons and deeper insights into the selection and integration of digital technologies.
2. **Digital technology initiatives:** To enable the empirical investigation of selection and integration processes, the selected cases needed to involve ongoing or recent projects in which digital technologies were being evaluated, selected, or integrated into the production system.
3. **Lean-based production philosophy:** As the selection and integration of digital technologies for smart production presuppose an understanding of value flows, processes, and operational challenges, the selected companies were required to actively work according to Lean principles. This consideration ensured a common baseline regarding process orientation and continuous improvement practices.

4. **Relevance to sustainability objectives:** The selected cases were required to address sustainability objectives from an economic, environmental, and/or social perspective. This characteristic had to be considered through formally articulated sustainability strategies or through digital technology initiatives that targeted efficiency, resource use, working conditions, or other sustainability-related outcomes.

Together, these four criteria resulted in a set of cases that represent diverse production system contexts, enabling a nuanced analysis of how digital technologies can be selected and integrated. Table 2 summarizes the company characteristics and provides an overview of the naming of the cases in the thesis and across the appended papers. The mapping clarifies how the cases referred to in the thesis correspond to those presented in the individual papers.

Table 2: Overview of the selected companies, cases studied, and their naming in thesis and appended papers.

	Company A		**Company B**	**Company C**	**Company D**
Company type	OEM		OEM	Supplier	Supplier
Product type	Heavy vehicles		Railway	Casted components	Electronics
Company size	Large		Large	SME	SME
Multi-/single site	Multiple sites		Multiple sites	Single site	Single site
Naming in thesis	Case A1	Case A2	Case B1	Case C1	Case D1
Naming in paper I	Case A	-	Case B	Case C	Case D
Naming in paper II	Case A	-	Case B	Case C	Case D
Naming in paper III	Case A	-	Case B	Case C	Case D
Naming in paper IV	Case 2	Case 1	-	-	-
Naming in paper VI	Case	-	-	-	-

3.3.3 Unit of Analysis

The unit of analysis refers to the main focus of inquiry within a case study (Yin, 2018). This thesis is positioned within the research area of the selection and integration of digital technologies in the context of the digitalization of

production. Accordingly, the primary unit of analysis is the selection and integration of digital technologies. However, the selection and integration of digital technologies is inherently complex and cannot be fully understood through a single analytical lens. To capture this complexity, embedded units of analysis were applied within the case studies. As shown in Table 3, different embedded units of analysis were applied across the case studies. This variation reflects the evolving research focus of the thesis and the need to explore the phenomenon from multiple analytical perspectives.

Table 3: Overview of context, unit of analysis, and embedded unit of analysis in the different case studies

	Case study Evoque	**Case study Velar**	**Case study Sport**
Case(s) included in the case study	A1, B1, C1, D1	A1, A2	A1
Context	Digitalization in production systems	Digitalization in production systems	Digitalization in production systems
Unit of analysis	Selection and integration of digital technologies	Selection and integration of digital technologies	Selection and integration of digital technologies
Embedded unit of analysis	Socio-technical aspects in smart production	Contextual complexity in smart production	Tensions in smart production projects
Presented in paper(s)	I, II, III	IV	VI

3.4 Research Process

The research process for each case study (i.e., Evoque, Velar, and Sport) involved interactions between existing academic literature and manufacturing practice, meaning that the research process was influenced by rising or dissolving opportunities at manufacturing companies (Karlsson, 2016). For this reason, it was considered essential to follow a clear research process involving different steps to help avoid failed research results (Yin, 2018). Each case study followed the research process steps that Säfsten and Gustavsson (2020) suggested, which Figure 4 illustrates. First, the problem area was defined by conducting a literature review to identify research gaps and having a dialogue with managers at the case companies to develop an understanding of challenges faced in practice. Second, based on identified research gaps and industry challenges, the purpose and research questions were formulated. The third step was to deepen the literature review to identify relevant literature for designing and conducting the case study. The knowledge gained from the literature review was used in the fourth step, which involved designing the case study. This step included identifying relevant

cases and planning how data should be collected. The case study design was conducted based on discussions and feedback from managers at the case companies. The fifth step involved collecting empirical data, such as interviews. The data collection is further explained in Section 3.5. The sixth step involved summarizing the empirical results and reflecting the overall impressions of the data collected by conducting discussions with managers and other researchers to ensure that enough data had been collected. The data were then analyzed and synthesized in the seventh step. A further description of the data analysis process is presented in Section 3.6. Before drawing conclusions from the conducted analysis, the results were discussed with managers at the case companies and other researchers to ensure a correct understanding of the analyzed phenomena. The final step involved discussing the results and drawing conclusions together with other researchers and industry managers.

As Karlsson (2016) explained, a research process has iterative steps and backloops. For this reason, these steps were not followed linearly. Instead, the research process was characterized by reflection and questioning arising from empirical findings.

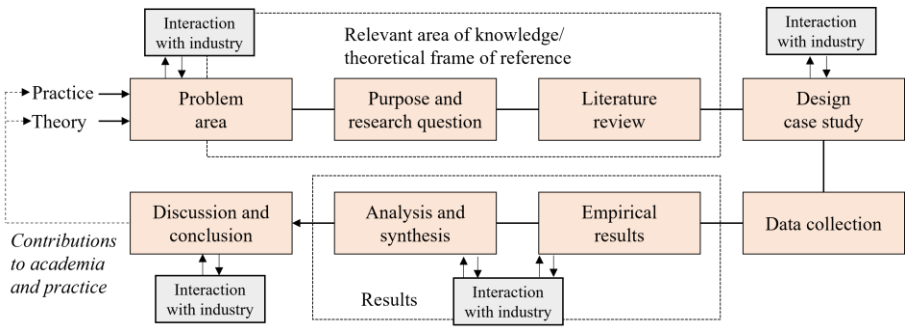


Figure 4: Research process applied for each case study, adapted from Säfsten and Gustavsson (2020)

The research process for the thesis involved analyzing the results from the licentiate defense and conducting an initial literature review to support the formation of new research questions. Each case study then followed the research process presented in Figure 4. Based on the formulated research questions, a synthesis of findings from the six appended papers was made, which is presented in this thesis.

3.5 Data Collection

This section describes how empirical data were collected within the three case studies Evoque, Velar, and Sport. It also describes the continuous literature review throughout the PhD studies, which is presented at the end of this section.

3.5.1 Case Study Evoque

In the Evoque case study (presented in Papers I–III), four cases were studied (A1, B1, C1, and D1), representing different organizational contexts and roles in digitalization initiatives. Case A1 and Case D1 served as the main cases, while Cases B1 and C1 were included as reference cases to complement and contrast the findings. This combination enabled both in-depth analysis and comparison across different contexts.

The unit of analysis was the selection and integration of digital technologies. To capture the complexity of this process, an embedded unit of analysis focusing on socio-technical aspects in smart production was applied. This focus made it possible to study the interaction among people, processes, and technologies in digitalization initiatives, providing a foundational understanding of key factors influencing selection and integration.

Data were collected using multiple qualitative methods, including semi-structured interviews, workshops, shopfloor observations, informal conversations, meetings, and document studies. These methods allowed for data triangulation and the inclusion of both strategic and operative perspectives. While the overall data collection approach was consistent across cases, the extent and intensity of data collection varied depending on the role of each case (main or reference case) and access to the organization.

Case A1 and Case D1 generated the most empirical data, including multiple interviews with different functional roles, workshops to explore and validate findings, and repeated shopfloor observations to capture production practices and the use of digital technologies over time. In addition, informal conversations and meetings were used continuously throughout the cases to clarify findings and support interpretation. Cases B1 and C1 were used as reference cases, with a more limited data collection focusing primarily on interviews, supporting documents, and single shopfloor observations. These cases provided complementary insights and enabled comparisons with the main cases.

A detailed overview of the data collection for each case, including the number and type of data sources, is provided in Table 4.

Table 4: Overview of data collected in case study Evoque

Technique	Case A1	Case B1	Case C1	Case D1
Interviews	Paper 1 and 2: 11 interviews × 30–80 min. Paper 3: 28 interviews × 30–80min. Head of global smart production, System and process developer, Head of IT, Smart production team leader	3 interviews × 47–66 min; Production manager, Production engineer	3 interviews × 47-66 min; Production engineer, Production planner, Production manager	23 interviews × 10–130 min; 6 operators, 2 Production engineers, Production manager, CEO
Workshops	Paper 1 and 2: 2 workshops × 90–120 min. Paper 3: 4 workshops × 60–120 min. Head of global smart production, System and process developer, Head of IT, Smart production team leader	-	-	5 workshops × 60-105 min; operators, Production engineers, Production manager, production planner, sales manager, quality manager, CEO, suppliers
Shop floor observations	Paper 1 and 2: 2 observations × 90–120 min. Paper 3: daily observations* × 10–120 min	1 observation × 60 min	1 observation × 60 min	Daily observations* × 10–120 min
Informal conversations	10 conversations × 5–15 min	2 conversations × 5–10 min	5 conversations × 5–15 min	Daily conversations * × 5–20 min
Various meetings	19 meetings× 10–60 min; Project meetings, meetings with software provider	-	-	38 meetings × 10–90 min; results, potential improvements, software applications/databases, long-term objectives and company performance
Documents	7 documents; Information about the company, available data, data available, data sources, software applications, digital technologies	3 documents; Information about the case company and the c	4 documents; Information about the case company and the case	36 documents; Information about the company, available data, available data sources, software applications, digital technologies.

* Observations and informal conversations were held daily when present at the case company (1-5 times per week)

3.5.2 Case Study Velar

In the Velar case study (Paper IV), two cases were studied within Company A, focusing on different application areas with varying levels of contextual complexity. One case addressed a remanufacturing ecosystem, representing a high-complexity context, while the other focused on performance monitoring using the IIoT across production sites, representing a lower to medium complexity setting. This design allowed for a comparative analysis of how contextual conditions influence the selection and integration of digital technologies.

The unit of analysis was the selection and integration of digital technologies. To capture how the selection and integration processes were shaped by the surrounding environment, an embedded unit of analysis focusing on contextual complexity in smart production was applied. This approach made it possible to analyze how factors such as organizational structure, stakeholder environment, system heterogeneity, and data uncertainty influence selection and integration processes.

Data were collected using semi-structured interviews and workshops, complemented by internal documents and company materials. The data collection was designed to capture both strategic and operational perspectives, as well as different functional roles involved in the digitalization initiatives. In total, eight interviews and four workshops were conducted in the remanufacturing case, while 12 interviews and two workshops were conducted in the performance-monitoring case. In addition, close interactions with managers were held to plan data collection and discuss findings.

A detailed overview of the data collection for each case is provided in Table 5.

Table 5: Overview of data collection in case study Velar.

Technique	Case A1 Data collected and Participants	Case A2 Data collected and Participants
Interviews	12 interviews × 30–80 min; Head of global smart production, system/ process developer, Head of IT, smart production team leader, operators, production managers, engineers	8 interviews× 30–60 min; Senior sales representative, business development manager, principal environmental lead, director of resource sustainability, circular product manager, engineers, senior data analyst
Workshops	2 workshops × 90–120 min; Head of global smart production, system/process developer, Head of IT, smart production team leaders, engineers	4 workshops × 90–120 min; Representatives from business development, circular economy, sustainability, engineering, manufacturing, production planning, digital manufacturing and data analytics
Shop floor observations	Daily observations × 90–120 min*	1 observation × 45 min
Informal conversations	Daily conversations × 5–15 min;* case-related information and feedback	Daily conversation × 5–15 min;* case-related information and feedback
Various meetings	25 meetings × 10–60 min; project meetings, software-provider meetings, and interactions with managers concerning case-study planning and results	12 meetings × 10–90 min; interactions with managers concerning case-study planning and results
Documents	10; Information about the company, digitalization strategy, available data, software applications, digital technologies	5; Information about the company, sustainability strategy, availabl data, software applications, digital technologies

*Observations and informal conversations were held daily when present at the case company (1-4 times per week) starting in 2024

3.5.3 Case Study Sport

In the Sport case study (Paper VI), a longitudinal single case study was conducted within Company A. The study followed the selection and integration of an IIoT technology over a period of approximately 5 years. The case included both retrospective and real-time elements, enabling the analysis of past decisions as well as ongoing digitalization initiatives. In addition, the case had a IMN perspective, including multiple production sites.

The unit of analysis was the selection and integration of digital technologies. To capture the challenges that arise over time, an embedded unit of analysis focusing on tensions in smart production projects was applied. This approach made it possible to identify and analyze competing demands, trade-offs, and contradictions that emerge during the selection and integration process.

Data were collected using semi-structured interviews, workshops, observations, and document studies between 2021 and 2026. From the beginning of 2024, closer interactions were held with managers to plan data collection and discuss findings. The data collection covered both strategic and operative perspectives and involved multiple functional roles, such as managers, engineers, and operators. A detailed overview of the data collection for the case is provided in Table 6.

Table 6: Overview of data collection in case study Sport.

Technique	Case A1 – Data collected and Participants
Interviews	28 interviews × 30–80 min; Head of manufacturing engineering and research, System and process developer, Head of IT, Smart production team leader, Operators, Production Managers, Engineers
Workshops	4 workshops × 90–120 min; Head of global smart production, System and process developer, Head of IT, Smart production team leaders, Engineers
Shop floor observations	Daily observations × 10–120 min*
Informal conversations	Daily conversations × 5–15 min*; Information related to the case and feedback
Various meetings	35 meetings × 10–60 min; Project meetings, meetings with software provider, interactions with managers related to planning and results of the case study
Documents	36 documents; Information about the company, digitalization strategy, available data, data, available data sources, software applications, digital technologies.

*Observations and informal conversations were held daily when present at the case company (1-4 times per week) starting in 2024

3.5.4 Literature Review

A literature review was conducted throughout the PhD work to develop an understanding of existing theories, previous research, and the underlying research problem driving this thesis. Literature was also used to support the design of case studies and to compare and validate empirical findings. During the PhD work, two main iterative literature review phases were conducted: one before and one after the licentiate seminar. However, literature was continuously reviewed throughout the entire research process.

In the initial phase of the PhD work, the literature review focused on gaining an understanding of the research area related to the selection and integration of digital technologies and socio-technical systems. This phase supported the formulation of the research questions for the licentiate thesis and the design of the Evoque case study. During the Evoque case study, literature was used to compare and categorize empirical findings, with a focus on the data value chain.

After the licentiate seminar, the literature review was extended to include sustainability, contextual complexity, and tensions within the scope of the selection and integration of digital technologies for smart and sustainable production. This

phase supported the development of research questions for the PhD thesis and the design of the Velar and Sport case studies.

Both literature review phases were conducted using scientific databases available through the Mälardalen University library, including Scopus, Web of Science, and Google Scholar. Different combinations of keywords were used to identify relevant journal articles and conference papers. The most commonly used keywords included:

- Selection and integration of digital technologies: selection, implementation, adoption, integration, digital technology, Industry 4.0, Industry 5.0, smart technology
- Socio-technical systems in smart production: socio-technical systems, human, people, technology, digital technology
- Data value chain: data generation, data collection, data communication, data transmission, data storage, data analysis, data processing, data usage, data visualization, data application, data management
- Sustainability: economic sustainability, social sustainability, environmental sustainability, ecological sustainability, Industry 5.0, sustainable manufacturing
- Contextual complexity: contextual complexity, organizational context, production context, manufacturing context, system complexity
- Tensions: paradox, trade-offs, organizational tensions, contradictions, management strategies, mitigation, managing paradox

To keep track of new publications related to the research topic, literature was monitored using database alerts for selected keywords and authors.

3.6 Data Analysis

A literature review was conducted in tandem with collecting and documenting empirical data. The empirical data in each case study were analyzed using a thematic data analysis, which is a systematic approach for capturing patterns in qualitative datasets (Braun & Clarke, 2006). The data analysis followed the six activities (Braun et al., 2019) described and involved iterations between empirical data and literature. Figure 5 depicts the data analysis process.

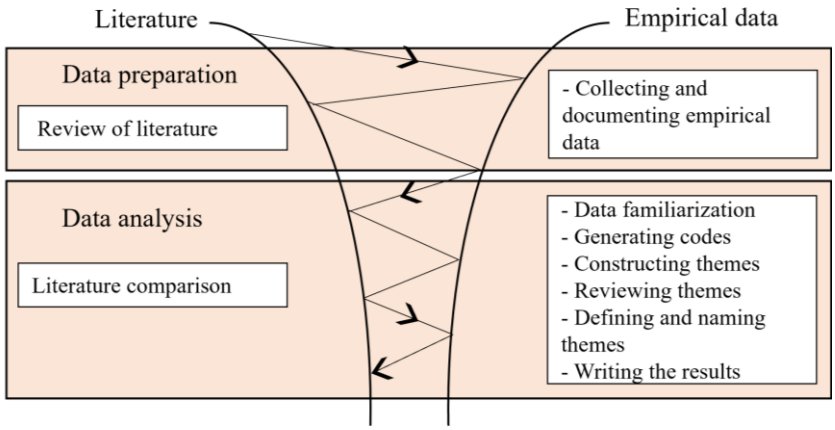


Figure 5: The iterative data analysis process in this thesis was based on the steps by Braun et al. (2019)

The first activity, data familiarization, involved reading transcriptions and notes to develop an overall understanding of the empirical data that had been collected. The second activity involved generating initial codes by highlighting interesting quotes and assigning codes that describe their meaning. The highlighted quotes and their associated codes were then transferred into a spreadsheet. Similar codes were grouped into first-order categories. The third activity involved constructing preliminary themes by identifying patterns among the first-order categories. The fourth activity involved reviewing themes by analyzing their relationships and comparing them to the literature. The fifth activity involved a further analysis of each theme and their relationships, resulting in second-order themes and an overarching third-order dimension. An example of the progression from empirical quotations to first-order categories, second-order themes, and third-order dimensions is presented in Table 7.

The data analysis was iterative and included interactions with both research supervisors and case companies. Emerging themes were first discussed with research supervisors and thereafter with case companies. These interactions were used to challenge and refine the findings and to ensure relevance to practice. Feedback from these interactions resulted in further analysis of the data collected and the refinement of categories, themes, and their relationships. Finally, the sixth activity involved writing the results.

Table 7: Example of coding from direct quotes

Direct quotes	First-order category	Second-order themes	Third-order dimension
"... He said "Oh, this is very bad, something really wrong happened, this is catastrophic, the whole production will be wrong with this". And since I'm familiar with the issues, I went to production and said "Hey, did you miss typed 76 instead of 86?" – and that was the case." **Production Manager, Company C**	Human factor affects data collection	Manual work processes and reporting	Data sources and collection
"The big limitation is exactly what I mentioned earlier that we have to generate the data somewhere and it will be a manual data generation. So that is the big difficulty that we have, finding a way to do it so that it is not noticeable that we are generating data manually. As an example, we log a lot when we start up a product and end a product and it is done in paper form and a protocol that is then entered into a computer. And there are processes like that to generate data, but it's still an administrative process, that someone sits and fills in a form even if you've done it on a computer. So, there is a difficulty in finding a way to still, with the other technical aids we have, collect data and go to the next step in such a way as to eliminate the administrative part of the whole thing" **Production Engineer, Company B**	Generating data from manual work processes		

The data analysis for the PhD kappa extended the analysis conducted in the individual case studies by comparing and synthesizing the findings across the included papers. Comparing the findings led to the identification of recurring patterns, differences, and connections and their use to develop the overarching findings of the thesis and answer its research questions. The analysis also included iterations among the findings of individual papers and the literature presented in this thesis.

3.7 Quality of Research

As this research adopted a case study method, the quality of the research was primarily assessed using the concept of trustworthiness (Lincoln & Guba, 1985), which is suitable for studies involving qualitative data, context-dependent phenomena, and interpretive analyses. This consideration is in line with case study research, where internal validity, external validity, and reliability are commonly used to assess research quality (Yin, 2018). The concepts of credibility, transferability, dependability, and confirmability provide an alternative but complementary way of addressing issues related to internal validity, external validity, and reliability.

Credibility concerns the extent to which the findings provide a believable and well-founded interpretation of the studied phenomena and accurately reflect the perspectives of the participants (Lincoln & Guba, 1985). This understanding is closely related to internal validity (Leedy & Ormrod, 2005; Yin, 2018). In this

thesis, credibility was strengthened by using multiple case studies and iteratively comparing empirical findings with existing research on smart production. A frame of reference was used to support the development of codes for interpreting the empirical data, and patterns identified in previous studies were used as the basis for comparison.

In addition, data were collected from different organizational perspectives, including both strategic and operative levels, which made it possible to capture different views of the same phenomenon. The triangulation of data sources, including interviews, observations, and documents, was also used to strengthen the credibility of the findings. Reflections and notes were written shortly after interviews and observations to capture important aspects and support the interpretation process. Member checks were also conducted with involved participants to validate interpretations and ensure that the findings reflected their experiences.

Transferability concerns whether the findings can be relevant in other contexts (Lincoln & Guba, 1985). Rather than aiming for statistical generalization, transferability is achieved by providing sufficient contextual detail to enable others to assess the applicability of the findings to their own settings, which aligns with analytical generalization in case studies (Yin, 2018). In this research, transferability was supported by including cases from different organizational contexts, such as large companies and SMEs, as well as different industries and application areas. This variation made it possible to identify patterns across different types of settings. Furthermore, the combination of real-time and retrospective cases made it possible to study both ongoing processes and completed initiatives. This approach provided an understanding of how digital technology selection and integration develop over time; it also supported the identification of recurring patterns across different conditions.

Dependability concerns the stability, transparency, and consistency of the research process over time (Lincoln & Guba, 1985) and is in line with reliability in case study research (Yin, 2018). In this research, dependability was addressed by using case study protocols and case study databases for documenting data collection and analysis. The data analysis process is described in Section 3.6. The case study protocol and database also support the establishment of a clear chain of evidence, which is emphasized in case study research (Karlsson, 2016).

Confirmability concerns the extent to which the findings are based on empirical data rather than the researcher's own assumptions (Lincoln & Guba, 1985). In this research, confirmability was supported through the triangulation of empirical data sources, including interviews, observations, and documents, as well as through continuous comparison with existing theory. The research process and findings were also continuously discussed and reviewed in several interactions

with managers from the case companies, allowing for the validation of interpretations. The research was also subject to continuous peer review through seminars, conference papers, and journal publications, where both methods and findings were scrutinized by other researchers. This activity can be seen as a form of external review, reducing the risk of bias.

In addition, the researcher's role and potential influence on the research process were reflected on throughout the thesis to ensure that interpretations remained grounded in the empirical data.

4 Summary of Appended Papers

This section presents a summary of the six papers included in this thesis. For each paper, the title, research gap, purpose, methodology, findings, implications, and contribution to the thesis are presented.

The appended papers address different aspects of the research problem, and their combined findings provide the basis for answering the research questions. Table 8 provides an overview of how the six appended papers relate to the two research questions of the thesis.

Table 8: Overview of the six appended papers in relation to the two research questions

	Paper					
Thesis Research Question	**I**	**II**	**III**	**IV**	**V**	**VI**
RQ1: How does the context of the production system influence the sustainable selection and integration of digital technologies for smart production?	X	X	X	X	X	-
RQ2: How should the sustainable selection and integration of digital technologies be managed in a production system context?	x	x	X	x	x	X

X = strong contribution; x = supporting contribution; - = no contribution

4.1 Paper I

Title: Critical Factors for Selecting and Integrating Digital Technologies to Enable Smart Production: A Data Value Chain Perspective

Research Gap: Existing research provides limited guidance on how manufacturing companies can systematically address challenges associated with selecting and integrating digital technologies when building data value chains for smart production.

Purpose: To identify and analyze the critical factors of selecting and integrating digital technologies for efficiently benefiting data value chains for smart production.

Methodology: The study adopts a multiple case study design involving four manufacturing companies of varying sizes and industries, enabling the exploration of contextual differences influencing digital technology integration. Empirical data were analyzed using the data value chain as an analytical framework to structure and compare integration challenges across cases.

Findings: The study applies the data value chain to analyze two in-depth cases, allowing for the mapping of the current state, identification of integration challenges, and definition of desired future data value chain configurations. A mapping example is presented in Figure 6. The findings are structured across five phases: (i) data sources and collection, (ii) data communication, (iii) data processing and storage, (iv) data visualization and usage, and (v) general integration-related factors. Across these phases, 13 themes of critical factors were identified. The results highlight the importance of clearly defining objectives and requirements for each phase of the data value chain to support effective selection and integration of digital technologies

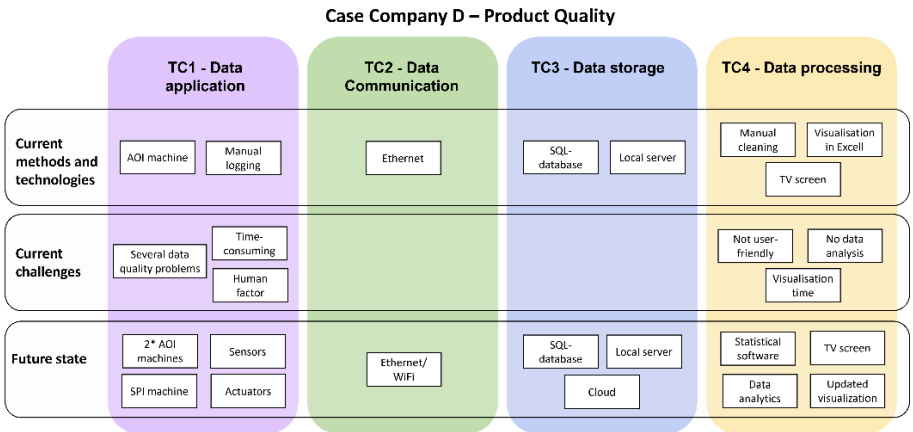


Figure 6: Mapping of technology clusters presented in paper 1

The study provides practical guidance for manufacturing companies by offering a structured approach for cross-functional collaboration among operational, IT, and engineering functions. The findings support managers and practitioners in identifying how to address challenges when selecting and integrating digital technologies to establish effective and value-creating data value chains.

Contribution to the Thesis: his paper contributes to answering Research Question 1 by analyzing two different production contexts: quality control in an SME and machine monitoring in a large manufacturing company. The findings show that differences in the existing data value chain and production conditions result in different challenges and requirements for digital technology selection and integration. The paper also contributes to Research Question 2 by demonstrating how mapping the current state and defining the desired future data value chain can support structured decision-making in digital technology selection and integration. The proposed mapping approach enables manufacturing companies to visualize existing technologies, methods, and associated challenges, providing a foundation for systematic selection and integration of digital technologies.

4.2 Paper II

Title: A Data Flow Framework to Support the Selection and Integration of Digital Technologies for Smart Production

Research Gap: Previous research has provided limited guidance on how manufacturing companies can systematically select and integrate combinations of digital technologies to support smart production. Existing approaches often focus on individual technologies or isolated parts of production systems whereas practical decisions are frequently based on experience and intuition rather than structured and holistic evaluation processes.

Purpose: The purpose of this paper is to support manufacturing companies in systematically selecting and integrating digital technologies to enable effective data value chains for smart production. To fulfill this purpose, the paper addresses two research questions:

1. What are the challenges of selecting and integrating a combination of digital technologies to build data value chains?
2. How can manufacturing companies systematically select and integrate combinations of digital technologies to enable smart production?

Methodology: A multiple case study involving four manufacturing companies of varying sizes and industries was conducted to capture different contextual conditions influencing digital technology selection and integration. The study builds on findings from Paper I and deepens the empirical and analytical understanding of technology selection and integration practices across different production contexts.

Findings: The study presents 15 challenges related to the selection and integration of digital technologies, which are categorized based on the data value chain. The help address these challenges, a holistic data flow framework structured around five technology clusters was proposed:

- TC1 – Data generation
- TC2 – Data communication
- TC3 – Data storage
- TC4 – Data processing
- TC5 – Data usage

Each technology cluster represents a set of digital technologies fulfilling specific functional roles within the data value chain. The framework emphasizes that all clusters need to be considered jointly to enable effective smart production; however, all clusters may not be relevant for all cases. Furthermore, the findings highlight the importance of maintaining an abstract and technology-agnostic cluster structure to ensure applicability across existing, emerging, and future digital technologies. The framework supports a systematic evaluation of alternative digital technology combinations and their relationships within the data value chain.

Figure 7 shows the framework based on the five phases of the data value chain, with arrows between them showing the possible directions of data flows between the phases. Data generation is the starting point. The in-depth analysis of the challenges in this paper contributes to a deeper understanding of the challenges manufacturing companies face when selecting and integrating digital technologies. The data flow framework represents one step toward overcoming these challenges as it shows how to systematically select, combine, and integrate relevant digital technologies for a specific application area in smart production.

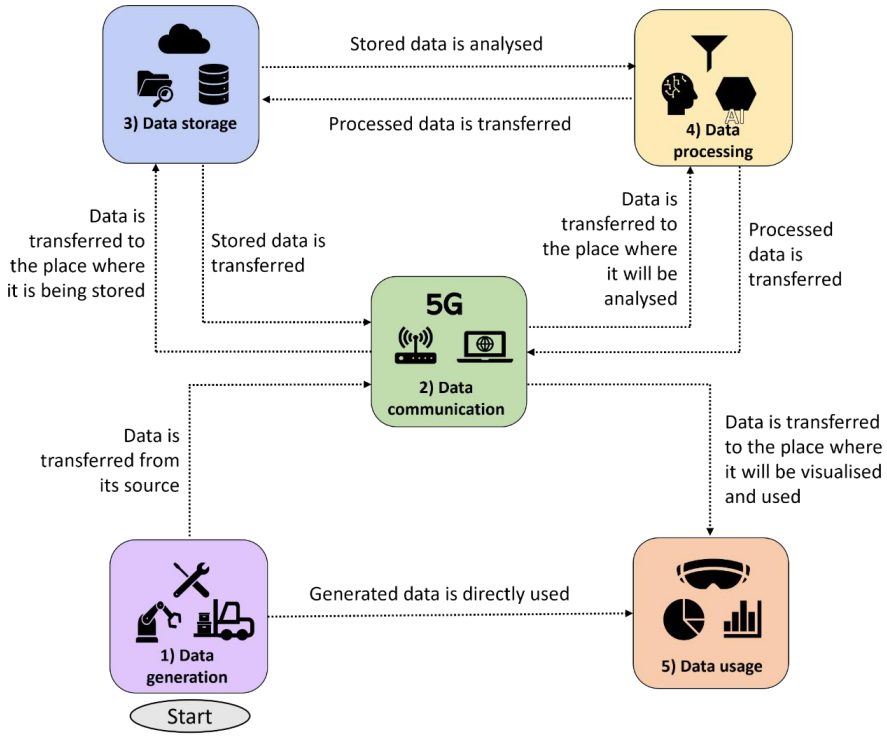


Figure 7: Proposed data flow framework to support the selection and integration of a combination of digital technologies (presented in paper 2)

Contribution to the Thesis: This paper contributes to Research Question 1 by structuring challenges encountered in digital technology selection and integration across the different phases of the data value chain. The findings show that requirements and challenges differ across data generation, communication, storage, processing, and usage, and that the relevance of different technology clusters depends on the specific application and production context.

The paper also contributes to Research Question 2 by providing a structured framework for systematically considering combinations of digital technologies and the data flows between them. The framework supports a holistic view of the data value chain, when digital technologies are not selected individually but considered in relation to existing technologies, desired benefits, and how they need to be integrated. It also supports cross-functional collaboration and the planning of digital technology integration activities, providing a foundation for systematic selection and integration of digital technologies.

4.3 Paper III

Title: Practices for Selecting and Integrating Digital Technologies Across the Data Value Chain

Research gap: Previous research on digital technology selection and integration has largely focused on technical aspects, such as technological capabilities, architectures, and technology evaluation frameworks. Although previous research increasingly recognizes the importance of socio-technical aspects, there is limited guidance on how these aspects can be translated into organizational practices for selecting and integrating digital technologies. More specifically, there is a need for more practice-oriented guidance on how manufacturing companies can select and integrate digital technologies across the data value chain to enable smart production.

Purpose: The purpose of this paper is to examine the practices that enable the selection and integration of digital technologies for smart production and to develop a process that supports the development of data value chains in manufacturing companies.

Methodology: A multiple case study was conducted involving four manufacturing companies of different sizes and industries. Data was collected through 57 semi-structured interviews, 12 workshops, and document analysis.

Findings: The findings revealed 16 organizational practices that should be considered when selecting and integrating digital technologies across the data value chain. The practices are organized according to data generation, communication, storage, processing, and usage. The practices address aspects such as defining data needs and quality expectations, engaging people in data collection, establishing interoperability, governing data access and security, defining storage and traceability needs, assuring data and analysis quality, accommodating different user needs, embedding data into organizational routines, and ensuring value realization.

Based on these practices, the paper proposes a process consisting of: Prepare, Select, Integrate, and Ensure Value. Prepare focuses on understanding the current state and defining the desired future state. Select concerns defining requirements and selecting digital technologies and an integration approach that can support the desired future state. Integrate focuses on embedding the selected digital technologies into the data value chain, work processes, and organizational routines. Ensure Value focuses on evaluating whether the intended value is realized and sustained, making improvements, and identifying further opportunities. The process is iterative rather than linear. Changes in needs, production

processes, and digital technologies can require existing data value chains to be reconsidered and improved over time. The process therefore connects Ensure Value back to an assessment of the current state and new improvement opportunities. A people, process, and technology perspective is considered throughout all four phases to support the alignment of digital technologies with organizational needs and existing production conditions.

Contribution to the thesis: This paper contributes to Research Question 1 by identifying practices that need to be considered when selecting and integrating digital technologies across the data value chain. The identified practices address different aspects of the production system, including existing equipment and systems, organizational structures, competencies, users, and data needs. The discussion further highlights that the relevance of the practices may differ depending on the production system context and that their application therefore needs to be adapted to the specific context.

The paper contributes to research question 2 by proposing a four-phase process for managing digital technology selection and integration. The process connects preparation, technology selection, integration, and ensure value rather than treating selection and integration as separate activities. The paper further contributes by showing the need for a holistic management perspective, where people, process, and technology are considered throughout the four phases. The process also shows that selection and integration need to be approached iteratively, as production needs, digital technologies, and data value chains can change over time and new learnings may require earlier decisions and practices to be reconsidered and adapted.

4.4 Paper IV

Title: Navigating Contextual Complexity in Smart and Sustainable Production: A Comparative Study on the Selection and Integration of Digital Technologies

Research gap: Although digital technologies are widely recognized as enablers of smart and sustainable production, existing research often treats selection and integration primarily as technical processes. Limited attention has been given to how contextual complexity, such as multi-actor environments, system heterogeneity, data uncertainty, and business model transformation, shapes how digital technologies should be selected and integrated. This limitation creates a need for empirical insights into how contextual conditions influence digital technology selection and integration strategies.

Purpose: The purpose of this paper is to examine how contextual complexity influences the selection and integration of digital technologies in smart and sustainable production.

Methodology: A multiple case study design was applied, examining two industrial cases within a global OEM in the heavy vehicle industry. One case focused on enabling a remanufacturing ecosystem (high contextual complexity); the other focused on performance monitoring across production sites (low-to-medium contextual complexity). Data were collected through semi-structured interviews, workshops, and document analyses. The empirical material was analyzed using a thematic analysis to identify dimensions of contextual complexity and corresponding integration strategies.

Findings: The study identifies seven dimensions influencing contextual complexity: process maturity, organizational landscape, stakeholder environment, system architecture, data uncertainty, integration demands, and transformation challenges. The findings reveal that contextual complexity significantly shapes priorities in digital technology selection and integration. In low-to-medium complexity environments, more emphasis should be placed on making well-informed technology selections early in the process, supported by standardized platforms, centralized governance, consistent data handling, and uniform key performance indicator (KPI) structures. In high-complexity environments, greater focus must be placed on integration strategies, including interoperability, federated governance models, distributed data architectures, and adaptable data-handling capabilities.

Based on these insights, the paper presents a structured framework outlining five key aspects to consider when selecting and integrating digital technologies according to contextual complexity: platform strategy, visualization, governance, data storage, and data-handling capabilities. An overview of these aspects is presented in Table 9.

The study advances theoretical understanding by demonstrating that the effective selection and integration of digital technologies must match the level of contextual complexity in the production environment. The findings show that more complex production systems require more adaptive and collaborative integration mechanisms. From a practical perspective, the framework provides context-sensitive guidance for manufacturing companies seeking to align smart production initiatives with operational conditions, stakeholder configurations, and sustainability ambitions. The findings support companies in balancing standardization and flexibility to enable scalable and sustainable digital integration.

Table 9: Overview of aspects to consider when selecting and integrating digital technologies depending on level of complexity (presented in paper IV)

		High complexity	Low-medium complexity
		With more complex, uncertain, and multi-actors in the context, more focus should lay on integration, not just choosing the technology.	The more structured, internal, and data-ready the environment, the more attention should be placed on making a suitable selection.
Selection	Platform strategy	A federated or loosely coupled data architecture may be better than a single platform due to many stakeholders with different systems.	A standardized platform (such as IIoT platform) is preferable as standardization improves speed, comparability, and maintenance.
	Visualisation	Role-specific or stakeholder-specific visualization tools tailored to the needs, language, and decisions of each group.	A standardized user-interface that can easily be rolled out on all sites (ready to be used) and/or an easy-to-use visualization tool for custom user-interfaces
	Governance	Requires strong, federated governance. Identify shared responsibilities and ownership models across stakeholders, ensuring selected technology can support distributed governance.	Internal governance with central support is sufficient. Use known roles, IT policies, and centralized access controls.
	Data storage	Distributed storage with life spam long retention, supporting traceability across systems.	Centralized storage aligned with internal MES/ERP systems, short- to medium-term retention.
	Data handling capabilities	Select technology that allows diverse data formats, asynchronous updates, and future interoperability.	Select technologies aligned with defined, internally consistent data requirements.
Integration	Platform strategy	Emphasize interoperability (APIs, shared schemas), not full platform unification.	Centralized IT/OT-led integration and support.
	Visualisation	Define different PIs and KPIs per role/function; allow tailored outputs for various stakeholders.	Central guidelines on how data should be visualized to enable unified PIs and KPI's.
	Governance	Needs strong data governance. Federated access rights, traceability protocols, audit trails.	Central integration support; sites maintain their systems independently, with shared definitions for reporting and KPIs.
	Data storage	Support distributed or centralized storage models with extended retention for life spam traceability of components or equipment.	Continue centralized storage model; retention typically short to medium term based on operational needs.
	Data handling capabilities	Enable flexible, cross-functional data interpretation across heterogeneous systems and stakeholders.	Integration must support consistent KPI logic, seamless connection to MES/ERP, and visualization tool. Prioritize internal consistency and comparability over flexibility and adaptability.

transformation initiatives with operational conditions, stakeholder configurations, and sustainability ambitions. The findings support companies in balancing standardization and flexibility to enable scalable and sustainable digital integration.

Contribution to the Thesis: Regarding research question 1, the paper shows that contextual complexity is an important aspect influencing the selection and integration of digital technologies. By identifying seven dimensions of contextual complexity and analyzing how they differ between the two cases, the study clarifies how the selection and integration of digital technologies must account for differences in process maturity, stakeholder environment, data uncertainty, integration demands, and transformation challenges. The findings further show that different levels of contextual complexity result in different priorities and requirements that need to be considered to achieve sustainable selection and integration of digital technologies.

Regarding research question 2, the paper contributes by developing a framework that supports manufacturing companies in adapting the selection and integration of digital technologies to different levels of contextual complexity. By distinguishing between low-to-medium and high complexity contexts, and by outlining aspects such as platform strategy, visualization, governance, data storage, and data handling capabilities, the study provides guidance for how digital technology selection and integration can be adapted to different production contexts. The paper thereby contributes to how sustainable selection and integration can be managed in relation to the complexity of the production context.

4.5 Paper V

Title: Towards Smart and Sustainable Production: A Conceptual Framework for Sustainability Considerations in Digital Technology Selection and Integration

Research Gap: Limited research has examined how sustainability aspects should be addressed when selecting and integrating digital technologies in manufacturing companies. Although digitalization and sustainability are often discussed together, existing studies rarely structure sustainability considerations across the data value chain or distinguish between sustainability aspects related to the selection and integration of digital technologies.

Purpose: The purpose of this paper is to identify and structure sustainability aspects that should be considered when selecting and integrating digital technologies to enable smart and sustainable production.

Methodology: The study is based on a literature review combined with workshops with industrial participants. Sustainability aspects were identified in previous research and discussed and refined together with practitioners. The findings were structured according to the phases of the data value chain. These phases were then analyzed in relation to economic, social, and environmental sustainability.

Findings: The findings show that sustainability aspects differ depending on the phase of the data value chain and whether decisions relate to the selection or integration of digital technologies. Economic aspects are often related to investment cost, lifecycle cost, reliability, and value creation potential. Social aspects relate to operator workload, safety, competence development, involvement, and acceptance. Environmental aspects relate to energy use, resource efficiency, technology lifetime, maintenance, and potential to support circular production practices. The results also show that sustainability considerations in upstream phases such as data generation and communication influence the ability to realize value in the data usage phase. The study presents a conceptual framework that links sustainability dimensions to the data value chain and distinguishes between aspects that are primarily relevant during selection and those that become critical during integration. The conceptual framework is presented in Table 10.

The framework provides structured support for manufacturing companies in discussing and evaluating sustainability when selecting and integrating digital technologies. It supports cross-functional dialogue between production, IT, and management by clarifying which sustainability aspects should be addressed at different stages of digital technology selection and integration. The framework can be used as analytical support to identify trade-offs and to align digital technology initiatives with sustainability targets and long-term strategy.

Table 10: Conceptual framework of economic, social, and environmental sustainability aspects to consider during the selection (s) and integration (i) of digital technologies across the data value chain (presented in paper V)

Phase of the data value chain	Economic Sustainability	Social Sustainability	Environmental Sustainability
Data generation	-Cost of technology (s) -Cost of integration (s,i) -Reliability and accuracy of data (s,i) -Lifecycle cost of technology (s) -Infrastructure readiness and legacy compatibility (s,i)	-Operator workload (s,i) -Safety and ergonomics (s,i) -Acceptance and ownership (s, i) -Skill requirements (s, i) -Feeling heard and involvment (s, i) -Avoidance of individual-identifiable data (s,i)	-Energy use of digital technologies (s) -Technology lifetime and recyclability (s) -Maintenance and upgrade needs (s, i) -Avoidance of unnecessary data generation (s,i) -Recyclability and reuse of hardware (s)
Data communication	-Cost of technology (s) -Maintenance cost (s,i) -Network reliability and uptime (s,i) -Scalability of communication solutions (s,i)	-Data ownership and access rights (i) -Transparency in communication layers (i) -Collaboration and trust between functions (s,i)	-Energy consumption of network infrastructure (s) -Flexible and modular system design for material efficiency (s,i) -Recyclability of hardware (s)
Data storage	-Total cost of ownership (cloud/on-premise) (s,i) -Value of data relative to storage cost (s,i) -Data retention vs storage cost (s,i) -Scalability and vendor lock-in risk (s) -Data security (s,i)	-Accessibility for employees (s,i) -Data governance and privacy (s,i) -Trust and ethical standards (s,i)	-Energy use of storage systems (s) -Data resolution, aggregation, and retention strategy (s,i) -Server hardware lifecycle and recyclability (s)
Data Processing	-Cost of technology (s) -Cost of integration (s,i) -Value in decision support (s,i) -Scalability of analytics and processing solutions (s,i)	-Human consideration and interpretability (s,i) -Analytical transparency and trust (s,i) -Cross-functional collaboration in analytics design (s,i)	-Energy consumption of computing (s) -Data filtering and resource optimization (s,i) -Processing location trade-offs (edge vs cloud) (s,i)
Data usage	-Cost of technology (s) -Cost of integration (s,i) -Productivity gains and cost reduction (s,i) -Long-term competitiveness (s,i) -Realized value vs expected value (s,i)	-Employee inclusion (s,i) -Skill development and data literacy (s,i) -Workload and cognitive ergonomics (s,i) -Collaboration and shared situational awareness (s,i) - Decision logic transparency and legitimacy (s,i)	-Energy consumption of visualization tools (s,i) -Recyclability and reuse of hardware (s) -Lifecycle thinking in digital technologies (s,i)

Contribution to the Thesis: This paper contributes to both Research Question 1 and Research Question 2. Regarding Research Question 1, the paper identifies and categorizes sustainability aspects that should be addressed when selecting and integrating digital technologies to achieve smart and sustainable production. By structuring the findings according to the phases of the data value chain and linking them to economic, social, and environmental sustainability, the study clarifies which sustainability aspects are relevant at different stages of digital technology deployment. The results show that sustainability considerations differ depending on whether decisions relate to selection or integration and that economic, social, and environmental aspects are interdependent rather than isolated. The findings also highlight that sustainability is closely connected to value realization in the data usage phase and that upstream decisions in data generation, communication, storage, and processing must be evaluated in relation to their contribution to intended decision support and improvement. From the thesis perspective, these sustainability aspects represent different priorities and conditions that need to be considered depending on the production system context and intended value. The paper thereby contributes to understanding what needs to be considered to achieve sustainable selection and integration of digital technologies.

Regarding Research Question 2, the paper contributes by proposing a conceptual framework that supports manufacturing companies in structuring sustainability considerations across both the selection and integration of digital technologies. By linking sustainability dimensions to the data value chain and distinguishing between selection-related and integration-related aspects, the framework provides a structured way to manage trade-offs and priorities in digital technology initiatives. The framework is not presented as a checklist, but as support to guide discussions and decision-making in practice. The paper thereby contributes to understanding how different considerations and priorities need to be addressed at different parts of the selection and integration process to support sustainable selection and integration of digital technologies.

4.6 Paper VI

Title: Managing Tensions in Sustainable Digital Technology Selection and Integration for Smart Production: An International Manufacturing Network Perspective

Research Gap: Previous research on digital technology selection and integration has mainly focused on critical success factors, barriers, capabilities, and technology-specific integration processes. Less attention has been given to the tensions

that arise when manufacturing companies need to address competing demands during selection and integration of digital technologies, especially across production sites within international manufacturing networks. There is also a limited understanding of how these tensions develop over time and how they can be managed as part of sustainable digital technology selection and integration to enable smart production.

Purpose: The purpose of this paper is twofold: (i) to analyse the tensions encountered during the selection and integration of digital technologies in IMNs and (ii) to develop a conceptual framework for managing these tensions to support sustainable selection and integration for smart production. To fulfil this purpose, two research questions were formulated:

- **RQ1:** What tensions are encountered during the selection and integration of digital technologies in IMN?
- **RQ2:** How can manufacturing companies manage the identified tensions to enable sustainable selection and integration of digital technologies in IMNs for smart production?

Methodology: The paper is based on a longitudinal single case study conducted within an international manufacturing network in the heavy vehicle industry. The research follows the selection and integration of an IIoT-based digital technology over approximately 5 years. Data were collected through semi-structured interviews, workshops, meetings, observations, and document studies. The case includes multiple production sites, allowing for the analysis of both real-time and retrospective data, while capturing variations across local contexts.

Findings: The findings identify five recurring tensions in the selection and integration of digital technologies: capability expansion versus time-efficient integration, global standardization versus local adaptation, access control and compliance versus speed and agility, champion-driven execution versus structured governance, and sustainability perspective versus technology performance prioritization. The tensions show that different demands can be important at the same time and that focusing on one demand can create consequences for the other. The tensions also reflect broader paradoxes related to organizing, performing, learning, and belonging.

Further, the findings show that these tensions cannot simply be resolved but need to be continuously managed. Four mitigation strategies are identified: acceptance, separation, integration of conflicting needs, and dynamic balancing over time. Which strategy is suitable can change as digital technology, production sites, and organizational conditions develop. Selection and integration of digital technologies are therefore an ongoing and iterative process in which

tensions need to be reconsidered over time rather than as a linear process ending when the digital technology has been integrated.

The paper provides a conceptual framework for supporting the management of tensions in sustainable selection and integration of digital technologies. The framework starts with the objectives of the digital technology initiative, followed by evaluating tensions, defining requirements, selecting suitable mitigation strategies, and selecting and integrating digital technology. Tensions are then continuously monitored as conditions change.

Contribution to the Thesis: This paper contributes to Research Question 2 by showing that managing sustainable selection and integration of digital technologies involves the continuous management of competing demands throughout the process. The paper adds a tension perspective to the thesis by showing that challenges during selection and integration cannot always be addressed by identifying one optimal solution, since different demands can be important simultaneously. The paper therefore contributes to understanding how competing demands can be identified, evaluated, mitigated, and continuously monitored as part of managing sustainable selection and integration. The identified mitigation strategies provide different ways of managing these demands depending on the production system context and how conditions develop over time. The paper also supports the view of selection and integration as interconnected and iterative, as decisions made during selection influence later integration. Learnings in integration can also create the need to revisit previous activities and decisions.

5 Analysis and Synthesis of Results

The purpose of this PhD thesis is to provide guidance to manage the sustainable selection and integration of digital technologies to enable smart production. The findings should empower manufacturing companies to orchestrate resources in a way that accelerates smart production. This chapter analyzes and synthesizes the findings from the papers appended to this thesis and answers the two research questions presented in Section 1.

5.1 Influence of Production System Context on Sustainable Selection and Integration of Digital Technologies

Papers I–V contribute different perspectives on how the production system context influences the selection and integration of digital technologies. The analysis for this thesis shows a set of recurring mechanisms related to the context of production systems that influence the sustainable selection and integration of digital technologies. These mechanisms include data value chain requirements, contextual conditions and priorities, and selection and integration practices. Previous research has already shown that smart production is dependent on context. For example, Demlehner and Laumer (2020) argued that digital transformation is strongly influenced by the characteristics of the production context, and Buer et al. (2020) examined how production environment and company size affect digitalization. The findings in this thesis build on this research by showing more specifically how and where the influence of the production system context becomes important during the selection and integration of digital technologies. The three identified mechanisms are analyzed in the following sections.

5.1.1 Data Value Chain Requirements

Previous research has highlighted that data are the key resource for supporting decisions to enable smart production, while digital technologies act as enablers that play different roles in managing and transforming raw data into useful insights that can be applied to add value (Miragliotta et al., 2018; Schuh et al.,

2020a). Moreover, previous research has argued that the different phases of the data value chain—comprising data generation, communication, storage, processing, and usage—must be considered in combination to generate value from data (Klingenberg et al., 2019; Tao et al., 2018). Indeed, more recent research has shown that smart production has mainly examined the standalone value of digital technologies, although their value commonly depends on integration with others (Varriale et al., 2025). This thesis builds on this perspective by highlighting how the production system context influences the requirements associated with each phase of the data value chain when selecting and integrating digital technologies. This also builds on previous research showing that different production application areas require different types of data and decision support depending on their purpose (Jardine et al., 2006; Juran, 1999; Reuter & Brambring, 2016; Valencia et al., 2019). The findings extend this by showing how these differences translate into requirements across the data value chain when selecting and integrating digital technologies.

Papers I and II demonstrate how the data value chain can be used to analyze the current and desired future state of a specific application area within a production system. By mapping current methods, technologies, and challenges, manufacturing companies can identify the requirements that digital technologies need to fulfill to support the desired future state. Mapping the data value chain also provides a common representation of the current situation, which can facilitate discussions among operators, production leaders, and technical specialists. Importantly, the analysis should include not only technical issues, but also organizational and soft factors, such as skills and work processes.

The cases show that the production system context has a strong influence on the requirements for data generation and communication. In the SME quality-control case, data were collected from both manual work activities and machine-generated sources. This context created requirements to either automate data collection or develop reliable manual processes together with operators. In addition, inaccurate machine-generated data demonstrated that improving existing data-generation processes could be more appropriate than selecting new technologies. In the multi-site performance-monitoring case, the need to collect data from heterogeneous and legacy production equipment placed greater importance on data communication. In contrast, the remanufacturing case required data generation from sources beyond the production system, including lifecycle data on used components, failure modes, and component condition history. Together, these cases show that data needs, work processes, location, and the heterogeneity of data sources influence the requirements for data generation and communication.

The production system context also affects the remaining phases of the data value chain. Data storage requirements depend on the volume and sensitivity of

the data, how long they must remain available, and who requires access. Performance monitoring involves large volumes of continuously generated machine data whereas remanufacturing requires selected lifecycle data to remain accessible over longer periods. Data-processing requirements depend on the structure and quality of the available data as well as the information needed to support decisions. Similarly, usage requirements vary according to the users involved and the decisions the data is intended to support, which supports previous research showing differences in data and visualization needs between production roles (Böck et al., 2025). More broadly, these finding are also in line with Eichenseer and Winkler (2024) work, which showed that data-oriented shopfloor management needs to be considered from technology, organization, and people perspectives rather than only from the availability of digital technologies.

Overall, the findings show that the production system context influences the requirements across the data value chain and, therefore, the suitability of different digital technologies as well. Paper V further shows that considerations in earlier phases of the data value chain need to be considered in relation to the intended value in the data usage phase. These findings are in line with previous research on technology interconnection and the complementary roles of digital technologies across the data value chain (Klingenberg et al., 2019). The findings in this thesis further show that the requirements for digital technologies and their combination depend on the particular production system context.

5.1.2 Contextual Priorities and Conditions

Previous research has increasingly considered economic, social, and environmental aspects in relation to digital technology initiatives and smart production (Martín-Gómez et al., 2024). At the same time, production system research shows that production contexts differ in their organizational structures, resources, competencies, responsibilities, and activities (Cheng & Farooq, 2018; Cragg et al., 2011; Nahm et al., 2003). Other studies have shown that digital technology initiatives are also influenced by contextual factors such as company size, available resources, technological intensity, competence, and existing infrastructure. For example, Buer et al. (2021) found that large manufacturing companies had higher levels of shopfloor digitalization and organizational IT competence than SMEs. They also noted that digitalizing a production system requires substantial expertise and investment because production systems are different, and there are few off-the-shelf solutions. More recent findings by Yli-Viitala and Helo (2026) similarly indicate that factors such as employee count and R&D spending are associated with smart production in manufacturing SMEs.

The findings in this thesis bring these perspectives together by showing that the production system context influences both what should be prioritized and the conditions required to realize these priorities, which is referred to as *contextual priorities and conditions*. Priorities concern what is considered important for a specific application; meanwhile, conditions concern circumstances that enable or constrain how the priorities can be addressed.

The defined priorities provide direction for what should be emphasized when selecting and integrating digital technologies. Paper V shows that these priorities can concern economic, social, and environmental sustainability, including lifecycle cost and value creation, employee workload and competence, and energy and resource efficiency. From the thesis perspective, these sustainability considerations represent different priorities that can influence what needs to be emphasized during selection and integration. Paper IV further shows how differences in contextual complexity create different conditions and priorities for digital technology selection and integration. The identified dimensions of contextual complexity, including process maturity, organizational landscape, stakeholder environment, system architecture, data uncertainty, integration demands, and transformation challenges, differed between the studied production system contexts. These differences resulted in different priorities and requirements for the selection and integration of digital technologies. This supports previous research showing that production system contexts can differ in processes, resources, competencies, responsibilities, and autonomy (Cheng & Farooq, 2018), while further showing how such contextual differences influence digital technology selection and integration.

However, these priorities are not necessarily equally emphasized across all application areas or production systems. The relative importance is influenced by the purpose of digital technology integration, organizational objectives, and external requirements. Buer et al. (2020), for example, found differences between SMEs and large enterprises in shopfloor digitalization and organizational IT competence while Yli-Viitala and Helo (2026) showed that firms differ in their ability to adopt technologies due to differences in learning and absorptive capacity. The findings in this thesis show that these conditions also influence what selection and integration alternatives are realistic. For example, Paper VI demonstrated that cases with limited competence can favor simpler digital technology or require competence development as part of the integration.

The organizational boundaries of the production system also create different conditions. Within a large manufacturing company, existing IT/OT structures, governance mechanisms, and central support can enable standardization across production sites while simultaneously constraining local technology choices. Indeed, research on international manufacturing networks has shown that

production sites can differ in their products, production processes, resources, competencies, responsibilities, and autonomy, creating a need for coordination and knowledge sharing across sites (Badasjane et al., 2022; Cheng & Farooq, 2018; Vereecke et al., 2006). For example, Pütz et al. (2023) acknowledged the need for heterogeneous production systems and processes when sharing learnings across sites.

The production system context therefore influences the sustainable selection and integration of digital technologies by affecting what should be prioritized and shaping the conditions under which these priorities can be realized. A priority does not necessarily align with the conditions available to address it. For example, environmental or social outcomes may be highly prioritized in an application area while limited financial resources constrain the digital technologies through which these outcomes can be achieved. Sustainable selection and integration therefore require priorities to be considered in relation to the conditions of the specific production system context.

5.1.3 Selection and Integration Practices

A third way in which the production system context influences the selection and integration of digital technologies concerns the *practices* through which these activities are performed. Previous research has identified several practices and factors important for smart production, including top-management commitment, employee involvement, training, cross-functional collaboration, and compatibility with existing IT infrastructure. For instance, Kassem et al. (2024) found that the integration of digital technologies requires the consideration of both technical and social aspects, including the existing IT infrastructure, management involvement, and operator training. Paper III extends this by identifying 16 practices across the data value chain that address different aspects of the production system, including existing equipment and systems, organizational structures, competencies, users, and data needs. The discussion in Paper III further highlights that the relevance of these practices may differ depending on the production system context and that their application therefore needs to be adapted to the specific context. The findings of this thesis support the importance of these practices while also highlighting which practices are needed and how they are performed and indicating that the resources required to perform them differ among production system contexts.

For example, the stakeholders who need to be involved depend on the organizational boundaries and complexity of the application area. In smaller companies or in a more bounded production context, the selection and integration of digital technologies may involve a limited number of people where responsibilities partly overlap. In a multi-site context, the same activities may require

collaboration among local production functions, IT and OT specialists, central functions, technology providers, and several production sites. The need for competence development also depends on existing internal knowledge. If the required competence is already available, digital technologies can be evaluated and integrated mainly using internal resources whereas other contexts may require support from suppliers or external specialists.

These examples show that the production system context influences not only which practices are important, but also how they need to be performed and which resources are needed to perform them. The evaluation of digital technologies can also differ depending on the context. In one context, the main need may be to evaluate whether a digital technology can fulfill a technical requirement; in another, the evaluation may also need to consider compatibility with existing systems, work processes, or user needs. This is consistent with previous research showing that smart production depends on both technical and organizational conditions rather than only on the digital technology itself (Zizic et al., 2022).

The findings also show that the extent to which existing practices can be reused differs according to production system contexts. In more established contexts, previous experience and existing knowledge can support selection and integration. Meanwhile, less familiar contexts may require different or additional practices. Overall, the production system context influences not only which selection and integration practices are relevant, but also how they need to be applied and the resources and stakeholders needed to perform them. Selection and integration practices should therefore not be applied uniformly but adapted to the specific production system context.

5.1.4 Research Question 1 Summary

The findings show that the production system context influences the sustainable selection and integration of digital technologies through three mechanisms: data value chain requirements, contextual priorities and conditions, and selection and integration practices. The context influences what is needed across the data value chain to support the intended use, what is important and feasible when selecting and integrating digital technologies, and how selection and integration practices should be carried out.

5.2 Managing Sustainable Digital Technology Selection and Integration

The analysis to answer Research Question 1 showed that the production system context influences the data value chain requirements, contextual conditions and priorities, and the practices needed for sustainable selection and integration of digital technologies. Research Question 2 builds on these findings by analyzing how these mechanisms can be managed during sustainable selection and integration of digital technologies. Paper III provides a starting point through a process for selecting and integrating digital technologies. The process includes activities related to understanding the current and future state, selecting suitable technologies, integrating them, and evaluating and maintaining the resulting data value chain. It also highlights the need to consider people, processes, and technology throughout these activities. However, Paper III also identifies context as an area requiring further research, particularly regarding how organizational and other contextual differences influence the selection and integration of digital technologies. The analysis for answering Research Question 2 therefore builds on the process presented in Paper III by considering how the production system context identified in Research Question 1 can be managed throughout the process. The analysis for answering Research Question 2 therefore builds on this process by synthesizing findings across the papers to develop a thesis-level process for managing sustainable selection and integration in a production system context. The individual papers provide complementary support for specific aspects of this process.

The process is structured into four phases: prepare, select, integrate, and ensure value. Although these phases provide an overall structure, the three mechanisms identified in Research Question 1 need to be considered throughout the phases. Thus, data value chain requirements, contextual conditions and priorities, and selection and integration practices are not considered at only one point in time. They have different implications depending on where in the selection and integration process the initiative is. The three mechanisms are also interrelated. Data value chain requirements define what is needed to support the intended value, while contextual priorities and conditions influence what is important and feasible. Together, these influence which selection and integration practices are needed and how they should be applied. At the same time, the practices used can provide new insights that lead to reconsideration of data value chain requirements, priorities, and conditions.

The analysis further showed that tensions, i.e. competing demands, need to be managed throughout the four phases. In addition, the process is not strictly linear, as new learnings gained during the phases can require earlier decisions to be

reconsidered and adapted. **Error! Reference source not found.**8 brings these findings together and presents the resulting process for managing sustainable selection and integration of digital technologies in a production system context.

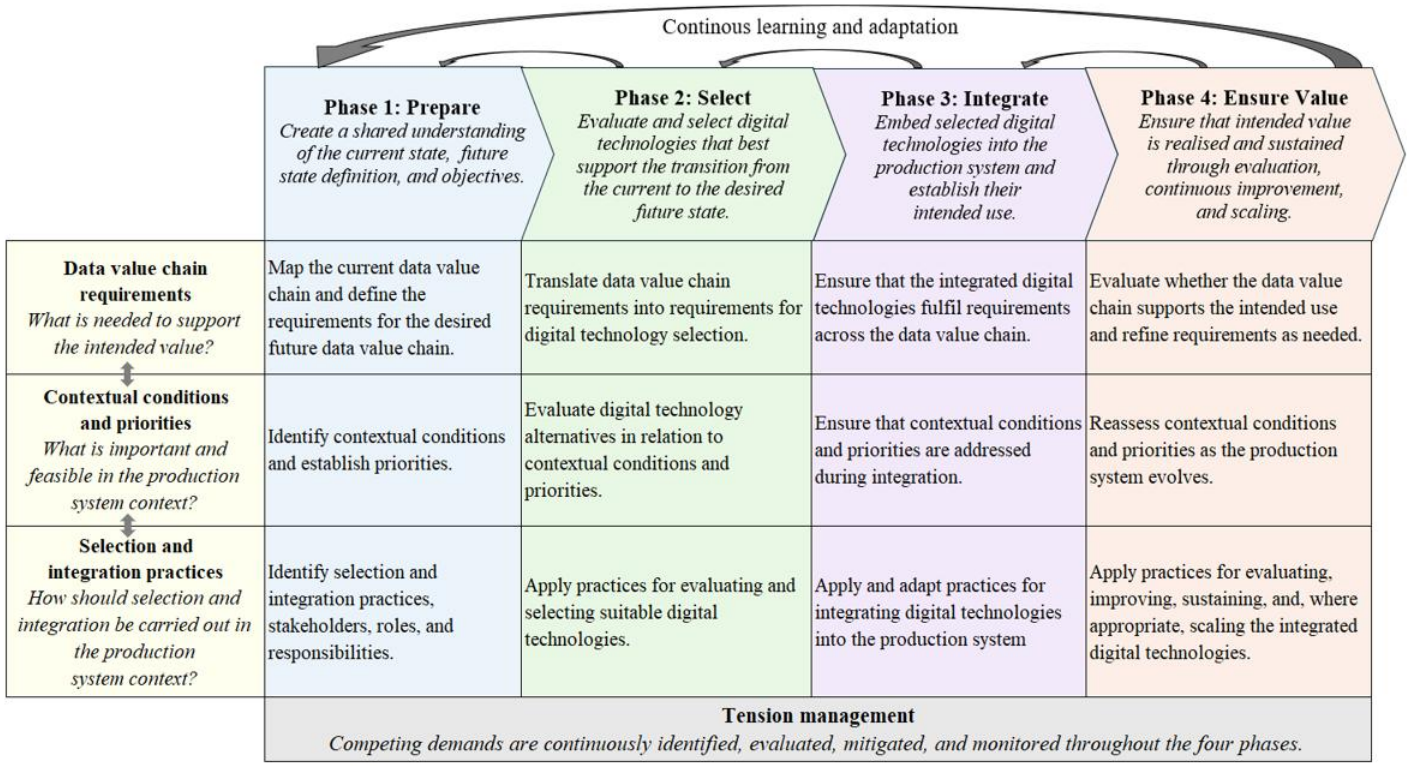


Figure 8: A process to manage the sustainable selection and integration of digital technologies in production system context

5.2.1 Phase 1: Prepare

The prepare phase concerns establishing a shared understanding of the current state, desired future state, and objectives. The findings from Papers I and II show that the data value chain can support this phase by making it evident how data are currently generated, communicated, stored, processed, and used as well as what needs to change to support the intended use. Paper II also shows that selecting technologies one at a time without considering the wider data value chain can result in incompatible technologies, interoperability problems, and investments that do not generate the expected value. The preparation should therefore start from what is needed from the data value chain rather than from a specific digital technology.

However, the findings from Research Question 1 show that defining requirements is not sufficient. The conditions of the production system also need to be understood. For example, the desired future state may require capabilities that are technically possible but difficult to achieve with the available equipment, competence, resources, or organizational structures. Paper IV supports this finding by showing that differences in contextual complexity create different requirements for coordination and integration. Digital technologies can also increase complexity through greater interconnectivity and autonomy, requiring updated integration strategies and coordination mechanisms. The prepare phase therefore needs to include an assessment of contextual conditions and priorities before deciding which technologies should be selected.

This preparation also influences how the selection and integration of digital technologies practices should be organized. Relevant stakeholders, responsibilities, and suitable practices need to be considered already during preparation. As Paper III details, digital technology integration involves interdependencies among people, processes, and technology, and these interdependencies differ depending on the activity and part of the data value chain. In a relatively bounded production context, this integration may involve a small group of stakeholders, while more complex initiatives can require coordination across production, IT, OT, different sites, or external actors. Preparing for selection and integration therefore means understanding not only what should be achieved, but also the conditions and resources available for achieving it.

The prepare phase thereby illustrates how the three mechanisms are connected. Data value chain requirements define what needs to be achieved, contextual conditions and priorities influence what is realistic and important, and the practices determine how the preparation needs to be carried out and by whom.

5.2.2 Phase 2: Select

During the select phase, the requirements identified during preparation are translated into more specific requirements for digital technologies. Paper II shows that smart production commonly requires combinations of interconnected technologies rather than one individual technology, yet no unique combination is suitable for all production systems. Thus, selection should consider both individual technical capabilities and how digital technologies can work together with the existing production system.

The contextual conditions and priorities identified during the prepare phase provide another basis for evaluating alternatives. A digital technology may fulfill the technical requirements but require the integration of resources, competences, or infrastructure that are not available. Another alternative may provide less functionality yet be easier to integrate and maintain. This approach differs somewhat from viewing digital technology selection mainly as identifying the digital technology that performs best according to a set of technical criteria. Rather, the findings indicate that suitability needs to be evaluated in relation to what should be achieved and the conditions under which it will be integrated. Paper VI similarly shows that selection decisions have implications for later integration and that digital technologies requiring major changes in processes or competence also create requirements for how integration needs to be managed.

The practices used for evaluating digital technology alternatives also need to reflect the production system's context. Paper III provides practices that support evaluation and selection, but the synthesis in Research Question 1 showed that the same practices are not necessarily needed or performed in the same way in every context. Testing may, for example, be more important where compatibility with existing equipment or systems is uncertain, while previous solutions can provide more support when the digital technologies and production context are already well understood. Therefore, the selection of digital technologies is not only about applying evaluation criteria, but also about finding ways of generating information to make the selection decision.

In the select phase, the three mechanisms therefore provide different but connected inputs to the selection decision. Data value chain requirements define what the digital technologies need to support, contextual conditions and priorities influence which alternatives are feasible and suitable, and selection practices influence how the necessary information is generated and evaluated.

5.2.3 Phase 3: Integrate

The integrate phase concerns embedding the selected technologies into the production system and establishing their intended use, which requires the requirements identified in the prepare phase to be realized in practice. Paper III shows that digital technology integration needs to consider people, processes, and technology rather than only technical implementation. Paper V similarly describes integration as embedding technologies into existing machines, systems, and work practices, which can involve technical connectivity, employee competence, process redesign, and cross-functional coordination.

The findings from Research Question 1 discussed in Section 5.1 further indicate that how this is done depends on the production system context. Integration with legacy production equipment, for example, requires different practices than integration into more recent and already connected equipment. Similarly, digital technologies that change how operators generate or use data require attention to competence and work processes, while integration across several production sites creates additional requirements related to coordination and governance. Integration practices therefore need to be adapted to the conditions identified earlier.

At the same time, not all conditions can be fully understood before the integration of digital technologies. In practice, integration can reveal limitations in data quality, compatibility, competence, responsibilities, or work processes that were not visible during preparation or selection. This distinction highlights one reason why selection and integration should not be managed as separate phases. Paper VI also argues that decisions during selection affect what is possible during integration and that integration problems can already originate from decisions made during selection. New learning can therefore create a need to reconsider earlier decisions.

Contextual conditions influence how digital technology can be integrated and which practices are needed, while integration activities can reveal new conditions or requirements that were not known earlier. These findings may require data value chain requirements, priorities, or practices to be reconsidered.

5.2.4 Phase 4: Ensure Value

The final phase, ensure value, concerns evaluating whether the integrated digital technologies actually support the intended use and whether this value can be sustained over time. This phase is important for sustainable selection and integration as defined in this thesis. A digital technology being only technically integrated does not necessarily mean that the intended value has been realized. Papers II

and VI, for example, examine cases in which technically functioning data value chains were still not used as intended and where the lack of user understanding and acceptance limited their longer-term value.

Evaluation therefore needs to consider the requirements defined in the prepare phase, including whether the data value chain provides the data required for the intended decisions as well as whether the contextual priorities have been achieved. Paper V shows that sustainability considerations occur at different points during the selection and integration of digital technologies and that the outcomes in data usage can be used to retrospectively evaluate earlier integrations in the data value chain. This finding suggests that evaluation should not only concern whether the digital technology works, but also whether earlier selection and integration practices actually resulted in the intended value.

The production system context can also change after integration. Production equipment can be replaced, competences develop, priorities change, and the digital technology may be scaled to another production area or site. Therefore, the contextual priorities, conditions, and requirements need to be reassessed. In this sense, scaling does not automatically mean copying an existing solution. The findings from Research Question 1 show that differences between production system contexts can require changes to both the digital technology and how it is integrated. Thus, the ensure value phase connects the outcome of one selection and integration initiative with further improvement and, when relevant, a new iteration of the process.

Ensure value therefore closes the loop between the three mechanisms. The achieved value is evaluated against the data value chain requirements and contextual priorities, while the results can show that conditions, requirements, or practices have changed or need further development. The phase can therefore provide input to a new Prepare phase rather than representing the end of the process.

5.2.5 Tension Management

Considering the three contextual aspects throughout the four phases does not mean that there will always be one solution that satisfies all requirements. Paper VI shows that digital technology selection and integration involve competing demands, including capability expansion versus time-efficient integration, global standardization versus local adaptation, control and compliance versus speed and agility, champion-driven execution versus structured governance, and sustainability considerations versus technology performance. These tensions are different from individual integration problems as both sides of the tension can be important. For example, standardization can support scaling and common

competence, while local adaptation can be needed to account for differences in machines, processes, and competence.

Tension management is therefore included across the four phases rather than as an additional phase. Paper VI identifies acceptance, separation, the integration of conflicting needs, and dynamic balancing as different ways of mitigating tensions. It also shows that the balance between competing demands can change over time as technologies and organizational contexts develop. Thus, a mitigation strategy used during selection may not necessarily remain suitable during integration or later use. Competing demands need to be identified and evaluated throughout the process, and the way they are mitigated needs to reflect the production system's context.

This finding also adds a new dimension to the structured process in Paper III. Paper III mainly provides activities for supporting the selection and integration of digital technologies; meanwhile, Paper VI shows that applying the phases does not remove the need for managerial support. Some decisions cannot be reduced to finding a perfect digital technology alternative, because fully addressing one demand can negatively affect another. Paper VI therefore shifts part of the management problem from solving individual challenges toward continuously managing competing demands.

The tensions can also arise through the relationships between the three mechanisms. For example, a desired data value chain requirement may conflict with available resources, or a priority for standardization may require practices that do not fit local production conditions. Tension management is therefore not separate from the mechanisms but concerns how conflicting requirements, priorities, conditions, and practices need to be managed throughout the process.

5.2.6 Continuous Learning and Adaptation

The findings also show that the four phases should not be managed as a strictly linear process. Paper VI describes the selection and integration of digital technologies as an ongoing process in which tensions develop over time and managers need to continuously adjust their approaches to competing demands. Paper IV also shows that digitalization can introduce new forms of complexity that require adaptive organizational structures and coordination mechanisms. Taken together with the findings from the other papers, this finding suggests that knowledge about both the digital technology itself and its compatibility with the production system context can be seen as an iterative development process.

This finding has different implications depending on how well the context is already understood. Where requirements, technologies, and organizational

structures are familiar, previous experience and existing practices can be reused to a greater extent. In more uncertain contexts, testing and practical integration can instead reveal requirements or limitations that were not known from the prepare phase. Earlier assumptions may then need to be reconsidered. The selection and integration of digital technologies can therefore also be understood as learning processes, rather than simply as a sequence of decisions and integration activities.

The feedback loops in the process represent this learning. Evaluation during the select phase can show that requirements established during the prepare phase need to be changed. Experiences during the integrate phase may require reconsideration of the selected digital technology or the integration approach. During the ensure value phase, actual use can show if further integration or changes to requirements are needed. Learning can therefore affect all three mechanisms. New learnings about contextual conditions and priorities can change what is required from the data value chain or which practices are suitable, while applying practices such as testing, evaluation, and stakeholder involvement can reveal previously unknown requirements or contextual conditions. Continuous learning and adaptation therefore connect the phases and allow for selection and integration of digital technologies to be adjusted as the understanding of the production system context develops and as the context itself changes.

5.2.7 Research Questions 2 Summary

Taken together, the analysis shows that managing sustainable selection and integration requires more than following the four phases identified in Paper III. The phases provide the structure for the process, while the findings from Research Question 1 provide three aspects that need to be managed within them: data value chain requirements, contextual conditions and priorities, and selection and integration practices. These mechanisms influence each other throughout the process. Contextual conditions and priorities influence what is required and feasible across the data value chain and how selection and integration practices need to be applied, while the application of these practices can reveal new requirements and contextual conditions. Paper VI further shows that competing demands need to be managed throughout the process, while the synthesis across the papers shows a need for learning and adaptation as knowledge about the production system context develops.

6 Discussion, Contributions, and Conclusions

This chapter discusses the results of the thesis, presents the contributions to theory and practice, and concludes with the main conclusions, limitations, and directions for future research.

6.1 Discussion

Previous research has established that selection and integration of digital technologies in production is context dependent. Differences related to company size, production environment, technological maturity, competence, and available resources have all been shown to influence digital technology adoption (Buer et al., 2021; Demlehner & Laumer, 2020; Yli-Viitala & Helo, 2026). The findings of this thesis support this view, but further indicate that considering contextual factors individually provides only a partial understanding of their influence on digital technology selection and integration. Production systems consist of interdependent elements, meaning that a condition that appears favorable from one perspective may create difficulties elsewhere in the system. As a result, context needs to be considered in relation to what digital technology is intended to achieve and how it fits with the existing production system. This also means that unfavorable contextual conditions should not necessarily be treated only as barriers that need to be overcome. Developing the competence, infrastructure, resources, or organizational conditions needed for an initiative can itself require substantial resources. These changes therefore need to be considered in relation to the value that the digital technology is expected to provide.

This idea becomes particularly important when digital technologies are selected. Previous research on selection has commonly evaluated technologies against predefined criteria, thereby providing a structured way of comparing alternatives. However, the findings in this thesis suggest that the criteria themselves cannot be assumed to have the same relevance across production systems. What constitutes a suitable digital technology depends on the data required for the intended use as well as the existing equipment and systems, competence, available resources, organizational structures, and priorities. This finding supports Buer et al.'s (2021) argument for a fit between digitalization and the production environment, but this thesis extends this argument to the selection and integration of

individual digital technology initiatives. However, fit with the existing production system should not necessarily mean that the current conditions are treated as fixed. Some conditions, such as competence or digital infrastructure, can be developed if the expected value justifies the required development effort. The suitability of an alternative therefore depends not only on the conditions that currently exist, but also on whether the conditions needed for its integration can reasonably be developed.

A related issue is that the suitability of digital technology cannot always be evaluated independently from other technologies. Previous research has increasingly recognized that value from digital technologies depends on their interconnection rather than only their individual capabilities (Klingenberg et al., 2019; Varriale et al., 2025). Varriale et al. (2025) pointed out that research has largely focused on the standalone value of technologies despite the importance of their integration. The findings of this thesis support this argument, but they also indicate that the required interconnection is also context dependent. Different intended uses create different requirements for how data needs to be generated, communicated, stored, processed, and used. This makes the evaluation of digital technologies more complex, as the suitability of one digital technology may depend on several other parts of the data value chain and the production system context. In practice, it may also be difficult to have sufficient knowledge about all these dependencies at the time of selection, particularly when different functions are responsible for different parts of the data value chain. Consequently, evaluating one digital technology in isolation may simplify the selection process, but it also risks overlooking conditions that later affect whether its intended value can be realized. Addressing this problem is, however, not straightforward, as considering these dependencies may require changes in both competence and ways of working. The conditions for doing so may also differ between companies. For example, SMEs may have shorter organizational distances between the actors involved and therefore greater possibilities for adapting how they work together, while larger companies may have access to more specialized competence and resources but face greater organizational complexity. This suggests that the challenge is not simply to consider more factors during selection, but to create the conditions needed to understand the relevant dependencies.

This conclusion also questions a clear separation between digital technology selection and integration. Selection research has primarily focused on how alternatives can be evaluated whereas integration research has focused more on barriers, critical success factors, and integration activities. In practice, however, decisions made during selection determine many of the conditions for later integration. Conversely, integration may reveal limitations that could not be fully understood during selection. Kassem et al. (2024), for example, highlighted that incompatibility with existing IT infrastructure can result in a newly implemented digital

tool becoming unusable. The findings in this thesis therefore suggest that selection and integration are better understood as interconnected rather than sequential problems. However, interconnecting selection and integration may be difficult in practice when responsibilities for these activities are separated within the organization. In an international manufacturing network, for example, digital technologies may be selected at a global level, while their integration is carried out by local IT and production functions across several sites. The actors involved in selection may therefore not have detailed knowledge of local production conditions, while those responsible for integration may have limited influence on earlier selection decisions. This creates a need to coordinate knowledge and decisions across organizational levels, functions, and sites. Such coordination depends not only on having a common understanding of how selection and integration should be approached, but also on involving actors with knowledge of local conditions early enough to influence selection decisions.

Another challenge concerns how selection and integration of digital technologies can become part of the daily work in production rather than activities carried out only within individual digitalization projects. Paper III showed that several of the identified practices can be performed as part of existing activities, for example when identifying needs, evaluating alternatives, integrating digital technologies, and following up their use. However, integrating these practices into daily work requires more than introducing a process. It also requires that the operators, production managers, engineers, and other actors involved have the competence, time, and responsibility to work with these questions and that this way of working is supported at different managerial levels. This may require changes in organizational culture, particularly if digital technology selection and integration have previously been treated mainly as technical or IT-related questions. Without such support, there is a risk that the practices are applied only when specifically required or depend on individual employees driving the work. This does not mean that all selection and integration activities can or should become routine activities, as some require dedicated expertise and cross-functional involvement. Rather, the findings indicate that parts of the process can be embedded in existing ways of working, while more specialized activities are supported when needed.

This thesis has implications for what sustainable selection and integration of digital technologies mean. A digital technology that appears suitable when selected may not generate its intended value after integration whereas a technical integration may lose value if the production system changes or the solution cannot be maintained. Sustainable selection and integration therefore cannot be determined only at the point of digital technology selection. It depends on whether the digital technology continues to fit the production system and whether the intended value can be realized and sustained over time. This finding is consistent with Franzè et

al. (2024), who show that sustained value from digital technologies depends on alignment with organizational capabilities and ways of working. The findings in this thesis extend this reasoning by connecting sustained value back to the way digital technologies are selected and integrated. However, sustainable selection of digital technologies also implies recognizing already during selection that the intended value cannot reasonably be achieved under the current conditions. For example, the required data value chain may not be affordable, or sufficient competence and resources may not be available to carry out and maintain the required integration. In such situations, continuing with the selection and integration process may consume resources without providing the intended value. Postponing or stopping the initiative until the necessary conditions can be developed may therefore be a more sustainable approach than proceeding with an alternative that is unlikely to provide sustained value.

At the same time, what constitutes a sustainable selection and integration of digital technologies is open to different interpretations. In this thesis, sustainable selection and integration primarily concerns whether the intended value can be realized and sustained over time. Other perspectives on sustainability may place greater emphasis on economic, social, and environmental aspects, either in how the selection and integration process is carried out or in the application area and effects of the digital technology itself. These perspectives are not necessarily in conflict, since economic, social, and environmental considerations can influence both the intended value and the conditions for realizing it. However, they may lead to different conclusions about which alternative, or whether any alternative, should be selected.

6.2 Contributions to Theory

This thesis contributes to theory by advancing the understanding of how production system context influences the sustainable selection and integration of digital technologies and how these activities can be managed to enable smart production.

First, the thesis contributes by conceptualizing the influence of production system context on digital technology selection and integration through three aspects: data value chain requirements, contextual priorities and conditions, and selection and integration practices. Previous research has shown that the suitability of digital technologies depends on their intended purpose and production system context (Martín-Gómez et al., 2024) and that smart production creates interdependencies among technologies, organizational structures, processes, and individual capabilities (Colombari et al., 2024). However, contextual aspects have commonly been examined individually, for example through technological,

organizational, or company-specific factors. The thesis addresses this fragmentation by providing an integrated conceptualization of how the production system context influences sustainable selection and integration. The three aspects provide a structure for analyzing what is required to support the intended use, what is important and feasible within the production system context, and which practices are needed to select and integrate digital technologies. This provides a basis for understanding context not as a set of isolated factors, but through their combined influence on digital technology selection and integration.

Second, the thesis contributes by connecting digital technology selection and integration as interrelated activities rather than treating them mainly as separate decision and integration problems. Previous research has predominantly approached selection through technology evaluation and decision-support methods (Aballay et al., 2023; Maretto et al., 2022; Santos et al., 2023), whereas integration research has focused more on determinants, barriers, critical success factors, and integration frameworks (Bueno et al., 2023; Ghobakhloo, 2019, 2020; Kassem et al., 2024). The thesis connects these research streams by conceptualizing selection and integration as parts of an interconnected process in which the conditions for integration form part of technology selection, while learnings developed during integration and use can inform subsequent decisions. This extends the understanding of selection beyond choosing a suitable digital technology and of integration beyond embedding an already selected technology into the production system.

Third, the thesis contributes to a context focused management process for sustainable selection and integration of digital technologies. Previous research has identified challenges related to coordinating digital technology initiatives across organizational functions and aligning strategic and operational requirements (Massaro Mendes da Silva & Albertin, 2024; Vaya-Arboledas et al., 2025), while limited guidance has been provided on how selection and integration should be managed together while accounting for production system context and changes over time. The process addresses this need by combining the four phases—prepare, select, integrate, and ensure value—with the three contextual aspects and mechanisms for tension management and continuous learning and adaptation. Rather than prescribing a fixed sequence or universally applicable solution, the process provides a structure for adapting how selection and integration are managed as the production system context, knowledge, and competing demands develop. In this way, sustainable selection and integration are conceptualized as an ongoing process aimed at realizing and sustaining the intended value rather than ending when a digital technology has been technically integrated.

6.3 Contributions to Practice

The thesis contributes to practice by providing a process for manufacturing companies to manage the sustainable selection and integration of digital technologies in different production system contexts. The process supports practitioners in evaluating digital technologies based on their intended purpose and production system context rather than primarily on the digital technology itself. By considering the production system context, practitioners can assess what is needed, what is important and feasible, and how selection and integration should be carried out. This can also help identify situations in which the conditions required to realize the intended value cannot reasonably be established, indicating that an initiative may need to be reconsidered or postponed.

The process also supports coordination across functions, organizational levels, and production sites by providing a common way of approaching selection and integration. This is particularly relevant when responsibilities are distributed among production, IT/OT, central functions, local sites, and technology providers. In addition, the process supports practitioners in identifying and managing competing demands rather than assuming that all requirements can be fully satisfied simultaneously.

Finally, the process supports selection and integration of digital technologies as continuous ways of working rather than isolated digitalization projects. Learnings gained during selection and integration can inform earlier and subsequent activities, while the ensure value phase encourages continued evaluation of whether the digital technologies generate their intended value. The process can thereby support manufacturing companies in progressively developing toward smart production.

6.4 Conclusions

Research Question 1 - How does the context of the production system influence the sustainable selection and integration of digital technologies for smart production?

The findings show that the production system context influences sustainable selection and integration of digital technologies through three mechanisms: data value chain requirements, contextual priorities and conditions, and selection and integration practices. The context affects which data and digital technologies are required for the intended use, what is important and feasible given existing priorities and conditions, and which practices are suitable for selecting and

integrating digital technologies. Differences in, for instance, production equipment, information systems, processes, people, organizational structures, resources, and application purposes therefore influence both the suitability of digital technologies and how their selection and integration need to be carried out. The production system context should consequently not be treated as a single external factor, but as something that shapes the selection and integration of digital technologies throughout the digital technology initiative.

Research Question 2 - How should the sustainable selection and integration of digital technologies be managed in a production system context?

The findings show that sustainable selection and integration should be managed through an iterative process based on the phases: prepare, select, integrate, and ensure value. Across these four phases, manufacturing companies need to consider data value chain requirements, contextual priorities and conditions, and selection and integration practices. These three mechanisms are interrelated and influence each other as the initiative develops. The process also requires continuous management of competing demands and continuous learning and adaptation as the understanding of production system context and digital technologies develops.

The process is therefore not linear. Selection may require earlier requirements to be reconsidered, integration may reveal limitations triggering selection decisions to be reconsidered, and evaluation of actual use may trigger further integration or a new iteration of the process. Sustainable selection and integration consequently involve continuously aligning the intended use, digital technologies, and production system context so that the intended value can be realized and sustained over time.

6.5 Limitations and Future Research

As with all research, the thesis has several limitations that should be considered when interpreting and applying the findings. First, the case study method is suitable for building a deeper understanding of how and why events occur as well as how they unfold over time. However, this method also implies limitations in terms of generalizability. The findings are based on a limited number of cases within specific production system contexts; thus, the results are not intended to be statistically generalizable. Instead, the thesis focuses on analytical generalization, where the findings contribute to theory and can be transferred to similar contexts. The proposed process should therefore be further evaluated in other production system contexts. Future research could apply and test the process in different manufacturing contexts, such as process industries characterized by

continuous flows and high levels of automation, highly regulated industries where traceability, data integrity, and compliance play a central role, and new production systems built from scratch. In addition, future research could benefit from other research methods, such as surveys, to statistically test and complement the findings identified in this thesis.

Second, the sustainability aspects identified in Paper V are subject to certain limitations. The literature review was not systematic, and the industry workshops involved a limited number of participants. As a result, the identified sustainability aspects should not be considered exhaustive. Additional economic, social, and environmental aspects may be relevant to the selection and integration of digital technologies, and their relative importance may vary across production system contexts. Future research could therefore conduct a systematic literature review and further empirical studies to develop and validate these aspects. This could provide more comprehensive support for considering sustainability priorities within the selection and integrations of digital technologies.

Finally, the proposed process represents a thesis-level synthesis of findings across the included studies and has not been empirically evaluated as a complete process. While the findings are grounded in the appended papers, future research should apply the complete process longitudinally in manufacturing companies to evaluate its usability and further develop the relationships between its phases and mechanisms.

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